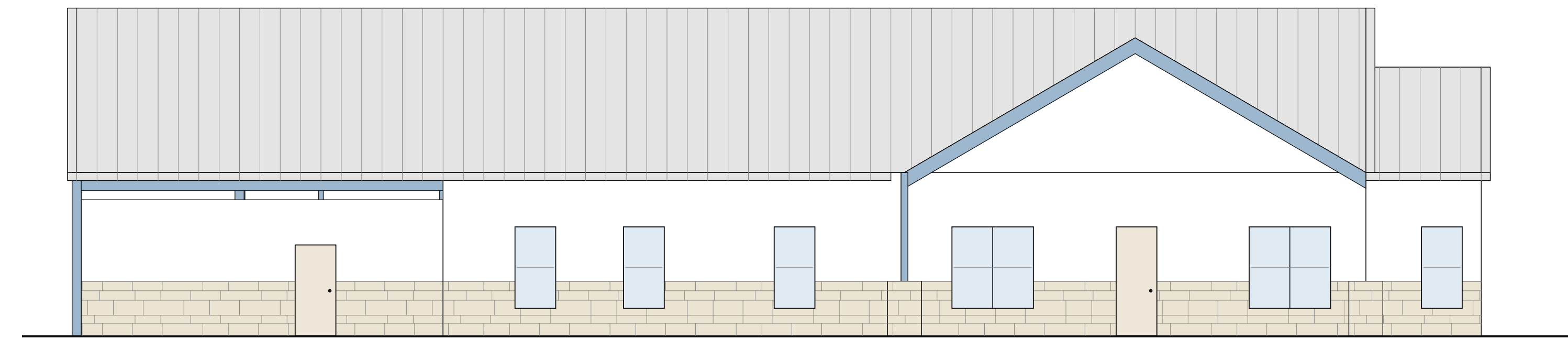


PEREZ RESIDENCE — PEREZ RANCH

Steel-frame / wood-wall hybrid residence · 43 Percy Howard Rd, Huntsville, Walker County, Texas

Preliminary design set for PE review and permit routing · 10/05/2026



North (front) elevation

PROJECT DATA

Owner	Carlos Perez
Site	43 Percy Howard Rd, Huntsville, TX 77320
Jurisdiction	Walker County, unincorporated (confirm outside Huntsville ETJ)
Use / occupancy	Single-family residence with attached in-law suite (IRC R-3)
Heated area	≈3,443 SF (main ≈2,663, suite ≈780)
Covered, unheated	Carport 864, breezeway 326, porches 1,066 SF
Structure	Steel rigid frames (HSS 8x8x3/8 / W14x22), 2x6 wood walls, 12' walls, 7/12 roofs
Wind	ASCE 7-22, 140 mph, Exp. C, partially enclosed
Code basis	Tex. Local Gov't Code §233.153 (IRC); inspections §233.154
Engineer of record	Texas PE — to be engaged
Room layout	Quality First Professional Design Services
Septic designer	K&B Tech, J. Katambani P.S. #3710, (936) 293-1598
Geotechnical	SoilsTEC, report KT26G0358

SHEET INDEX

G-001	Cover, sheet index and permit routing
G-002	Design criteria, structural notes and open items
C-101	Site plan
A-101	Floor plan
A-201	Exterior elevations — north and south
A-202	Exterior elevations — east and west
A-301	Building sections
A-501	Typical wall section and wall details
A-601	Door and window schedules, general notes
S-101	Foundation and anchor bolt plan
S-201	Wall, column and bracing plan
S-301	Roof framing plan
S-401	Frame elevations
S-501	Typical connection details
P-101	Plumbing coordination plan

DEFERRED SUBMITTALS (BY OTHERS)

- Windows and exterior doors: DP-50 ratings and installation instructions (owner-supplied).
- Standing seam roof: tested uplift system data (UL 580 Class 90 or FM 4471) on 1/2" plywood.
- PBR wall panels: maker's load tables and fastener schedule.
- Spray foam: product evaluation report.
- Lightning protection: NFPA 780 / UL 96A design.

PERMIT AND APPROVAL ROUTING

Approval	Agency	What to submit	Sheets
1. 911 address	Walker County Planning & Development (911 Addressing), 1313 University Ave, Huntsville · (936) 436-4939	Deed, parcel location. Allow 7-10 business days; pick up the address stake in person. Needed before any county permit.	C-101
2. Development permit (single family)	Walker County Planning & Development (same office)	911 address, recorded deed (vol./page), owner photo ID, site drawing with road, north arrow and two perpendicular measurements to scale, floor plan. Certificate of compliance if inside the Huntsville ETJ. Fee \$100 (verify).	C-101, A-101
3. OSSF (septic) permit	Walker County Planning & Development (OSSF)	K&B Tech design with the spray areas 25-30' + from any slab; aerobic system affidavit recorded with the County Clerk; maintenance contract. Fee 210 + 26 affidavit (verify).	C-101, A-101
4. Floodplain development	Walker County Floodplain Administrator (Planning & Development)	Only if the parcel is in or next to a mapped flood zone: FEMA panel, finished floor elevation, elevation certificate.	C-101, A-301
5. Driveway / culvert	Walker County Road & Bridge (Percy Howard Rd); TxDOT if access is from SH 19	Driveway location, culvert size and material.	C-101
6. Electric service	Electric utility serving the parcel	New-service application, meter location, load estimate.	C-101, A-101
7. Engineering seal	Texas licensed PE (structural)	Review, revise and seal S-sheets and A-301/A-501; calculations; foundation design from the geotech report.	All S- and A-sheets
8. Required inspections (\$233.154)	Licensed engineer, TREC inspector or ICC residential combination inspector	Foundation before the pour; framing and MEP before drywall; final. File results with the county within 10 days of the final.	Sealed set
9. Lender (construction / VA loan)	Lender and appraiser	Sealed plan set, specifications and cost breakdown.	Full set

PRELIMINARY
FOR PE REVIEW
NOT FOR
CONSTRUCTION

PEREZ RESIDENCE

Perez Ranch
43 Percy Howard Rd, Huntsville, TX
Walker County (unincorporated)

Owner: Carlos Perez

ENGINEER OF RECORD (TEXAS PE)



Name / firm / F-number:

REVISIONS

No.	Date	Description
0	10/05/2026	Preliminary for PE review
1	10/08/2026	Address; fireplace removed; vaults; BOM file renamed

PREPARED FROM

Owner's 3D frame model and BOM
(Perez Ranch Material BOM.xlsx).
Room layout: Quality First Professional
Design Services plan set (rev. 03-25-26).
Geotech: SoilsTEC KT26G0358.
Septic: K&B Tech OSSF plan, 08-14-26.

SHEET TITLE

COVER, SHEET INDEX
AND PERMIT ROUTING

DATE	DRAWN
10/05/2026	Owner / Claude

SHEET

G-001

Preliminary design. The Texas PE's sealed drawings and calculations govern.

DESIGN CRITERIA

Building code	Tex. Local Gov't Code §233.153; IRC as applied in Huntsville (or IRC as of May 1, 2008). Steel frame designed by a Texas PE.
Wind	ASCE 7-22, V = 140 mph ultimate (3-s gust), Exposure C, Risk Category II, Kzt = 1.0, Kd = 0.85, q = 37.6 psf at h ≈ 18'.
Enclosure	Whole house partially enclosed, GCpi = ±0.55 (open carport and breezeway, broken-window case, ASCE 7-22 tornado internal pressure).
Tornado	140 mph partially enclosed design = EF2 envelope.
Dead load	Roof ≈ 10.7 psf (24 ga panel, 1/2" plywood, underlayment, purlins, spray foam, 5 psf collateral, frame). Walls ≈ 6 psf (26 ga PBR, OSB, 2x6). Limestone wainscot ≈ 45 psf × 4' = 180 plf on the slab-edge ledge.
Live / snow	Roof live load 20 psf (governs over snow and ice).
Seismic	Seismic Design Category A.
Soils	SoilsTEC KT26G0358: CH clay Pl to B2; PVR 3.3" as-is, 1.0" with 4' select fill; bearing 1,400 psf sustained / 2,100 psf total.
Components	Windows and doors DP-50 (corner demand ≈ 42 psf ASD). Roof clips 24" o.c. field, 12" o.c. edge and corner zones (= 4'-2" bands).
Durability	100-year target: galvanized exposed steel, G90 cold-formed members, AZ55 Galvalume + PVDF roof (UL 2218 Class 4, UL 790 Class A), NFPA 780 lightning protection.

OWNER-SIDE CHECK SUMMARY (ASD utilization; PE calculations govern)

Element	Size	Util.	Notes
Roof deck	1/2" CDX plywood on purlins at 24"	0.41	Span rating 32"
Roof clips	2 × #10 screws per clip, 24" o.c.	0.65	12" o.c. in edge/corner bands
Roof purlins	8" Z 14 ga at 24" o.c.	0.95	Edge-zone uplift, 20'-6" bays
Main frame column	HSS 8x8x3/8	0.69	Knee moment governs
Main frame rafter	W14x22	0.72	Flange braces at every other purlin
Porch frames B, C	HSS 6x6x3/16, W12x14	0.48	
Suite frames	HSS 6x6x3/16, W12x14	0.58	
Carport header	W16x45, 27' span	0.37	Sized for a future attic floor (~0.8)
Grid 2 frame / tie rod	Column 2' in from grid B; 1" rod	0.35 / 0.45	Tie rod ~7.8 k tension
Grid 7 corner columns	HSS 6x6x3/16 to main rafter	0.52	Bedroom 3 bump corners; L/335
Grid 7 stub post	HSS 4x4x1/4 on bump rafter	0.07	Above bedroom 3 ceiling
Wall studs	2x6 DF #2 at 16", 12"	0.72	L/272
Eave strut	Span ≤ 10'-5"	0.57	Ceiling-height walls braced by the ceiling
Wind posts	HSS 4x4x1/4	0.60	
Gable girts	8" C 14 ga	0.62	Grid 7 spans ≤ 7' (0.29)
Portal frames	HSS 8x8x3/8 + W12x26	0.80	
Rod bracing	7/8" rod	0.83	
Anchor bolts	4 × 3/4" F1554 Gr 36	0.22	Uplift 8.3 k per main column

STRUCTURAL GENERAL NOTES

- Structural steel: HSS ASTM A500 Gr C; W shapes A992; plates, angles and rods A36; anchor rods F1554 Gr 36; bolts A325. Fabricate and erect per AISC 360 and the AISC Code of Standard Practice.
- Cold-formed purlins, girts and eave struts: AISI S100, G90 galvanized (ASTM A653). Lap purlins at frames per the manufacturer.
- Welding: AWS D1.1 by certified welders. Bolted moment connections preferred at knees and ridges.
- Exposed steel (carport, porches, header): hot-dip galvanized (ASTM A123) or zinc-rich primer + epoxy + urethane. Cap and seal HSS columns.
- Temporary bracing: erector to brace frames until purlins, roof deck and permanent bracing are complete.

OPEN ITEMS BEFORE CONSTRUCTION

Item	What's needed	Who
Foundation	PTI or BRAB slab from SoilsTEC values; select fill for ~1" PVR; hold-down for up to 8.3 k uplift per column; 13 drilled piers under the main columns (S-101)	PE
Stone wainscot	5" slab-edge ledge for 4' limestone (= 180 plf) around the house and stone piers at porch posts; veneer ties per TMS 402 at 140 mph	PE
Plumbing rough-in	Coordinate drains and the island conduit with beams and footings per P-101 (12 items); rough-in survey before the pour	Plumber / PE
Infill walls under ceilings	Carport and breezeway walls braced at the top by the 10' ceiling / attic floor diaphragm	PE
Back porch recess	Soffit or ceiling closing the strip between the 12' plate at the great room rear wall and the main roof (up to 14'-8") over the porch recess	PE
Portal bays	Wind posts at 55'-1" and 85'-8" bolt to the PF-1 beams; PF-2 at grid B laundry bay	PE
Roof overhangs	12" eaves and 8" rakes: overhang uplift in the C&C design, eave extensions, clip spacing	PE
Carport roof	Open-building net pressures under the main roof	PE
Vaulted great room	Valley framing, cut main purlins, grid 6 frame through the vault	PE
Attic space (study)	If used for a gym or rooms: floor at ~11' (~9' over the suite), steel floor beams over the carport, insulated floor over the open carport, stairs, frame check	PE
Site wind speed	ASCE Hazard Tool report at the house lat/long	Owner
Exposure and terrain	Confirm open terrain, no hill or escarpment (Exp. C, Kzt = 1)	Owner
Survey	Boundary, building stake-out, finished floor elevation	Surveyor
Flood	FEMA map lookup; floor ≥ 1' above BFE or 18" above grade	Owner
Septic spray field	Spray circles 25-30'+ from any slab	K&B Tech
Site address	One 911 address for all permits	Owner
Geotech reliance	Reliance letter naming owner and PE; SoilsTEC plan review and pier addendum	SoilsTEC
Freeze protection	Supply lines in interior walls, insulated	Plumber

- Concrete: 4,000 psi at 28 days; reinforcing ASTM A615 Gr 60; slab and beams per the PE's design from SoilsTEC KT26G0358.
- Wood: 2x6 Douglas fir #2 studs and plates, treated bottom plate (UC4A), 7/16" OSB (APA rated). Connectors hot-dip galvanized or stainless where in contact with treated wood.
- Roof deck: 1/2" CDX plywood (32/16), nailed per the PE as a diaphragm.
- Frame drift = 1.3" at 12' under design wind: provide slip joints at the top of partitions attached to frames.
- Owner-side checks on this set are preliminary. The Texas PE's sealed drawings and calculations govern.

PRELIMINARY

FOR PE REVIEW

NOT FOR

CONSTRUCTION

PEREZ RESIDENCE

Perez Ranch

43 Percy Howard Rd, Huntsville, TX

Walker County (unincorporated)

Owner: Carlos Perez

ENGINEER OF RECORD (TEXAS PE)



Name / firm / F-number:

REVISIONS

No.	Date	Description
0	10/05/2026	Preliminary for PE review
1	10/08/2026	Address; fireplace removed; vaults; BOM file renamed
2	10/08/2026	Drilled piers (S-101)

PREPARED FROM

Owner's 3D frame model and BOM

(Perez Ranch Material BOM.xlsx).

Room layout: Quality First Professional

Design Services plan set (rev. 03-25-26).

Geotech: SoilsTEC KT26G0358.

Septic: K&B Tech OSSF plan, 08-14-26.

SHEET TITLE

DESIGN CRITERIA,
STRUCTURAL NOTES
AND OPEN ITEMS

DATE	DRAWN
10/05/2026	Owner / Claude

SHEET

G-002

Preliminary design. The Texas PE's sealed drawings and calculations govern.

1

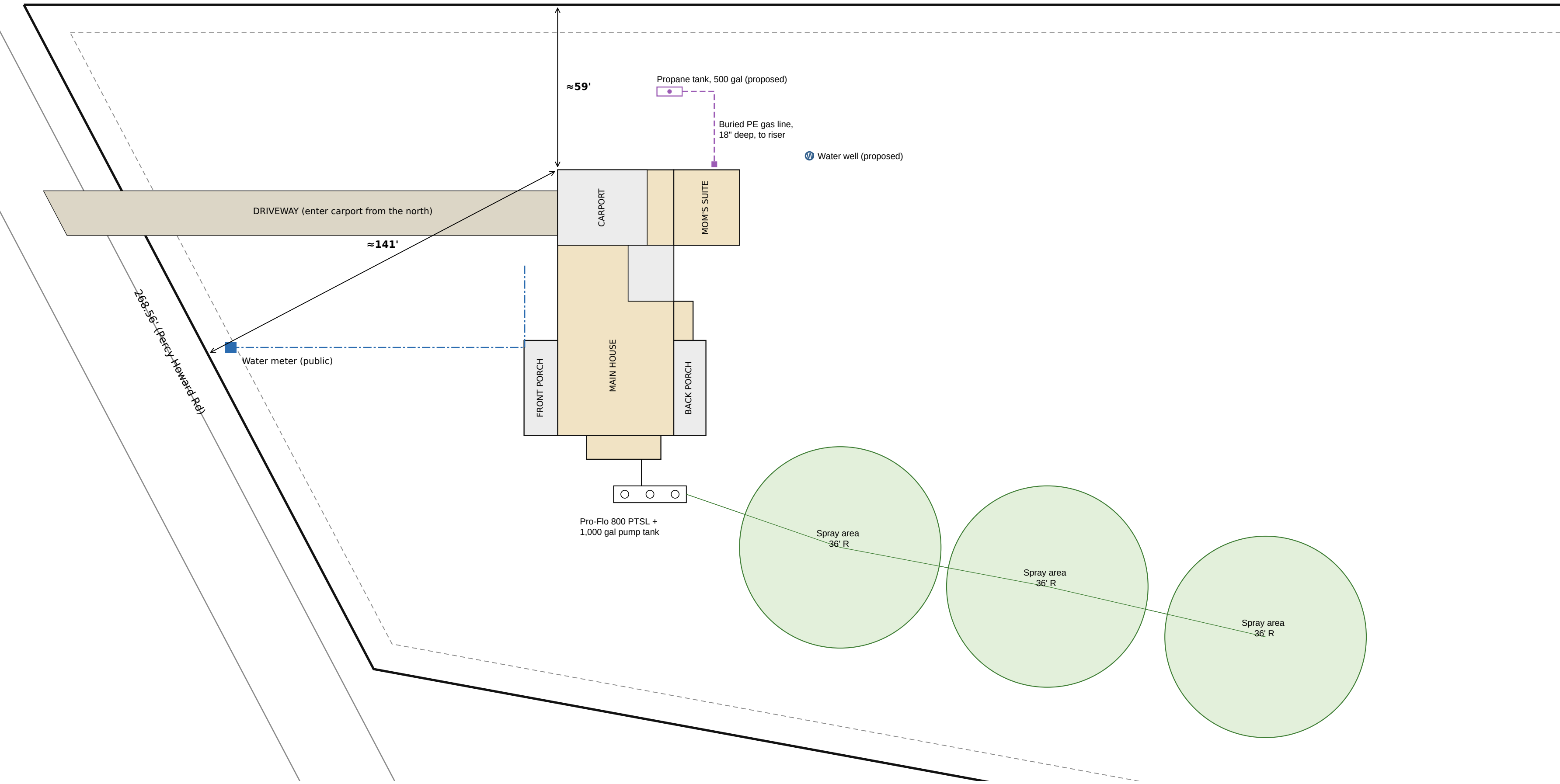
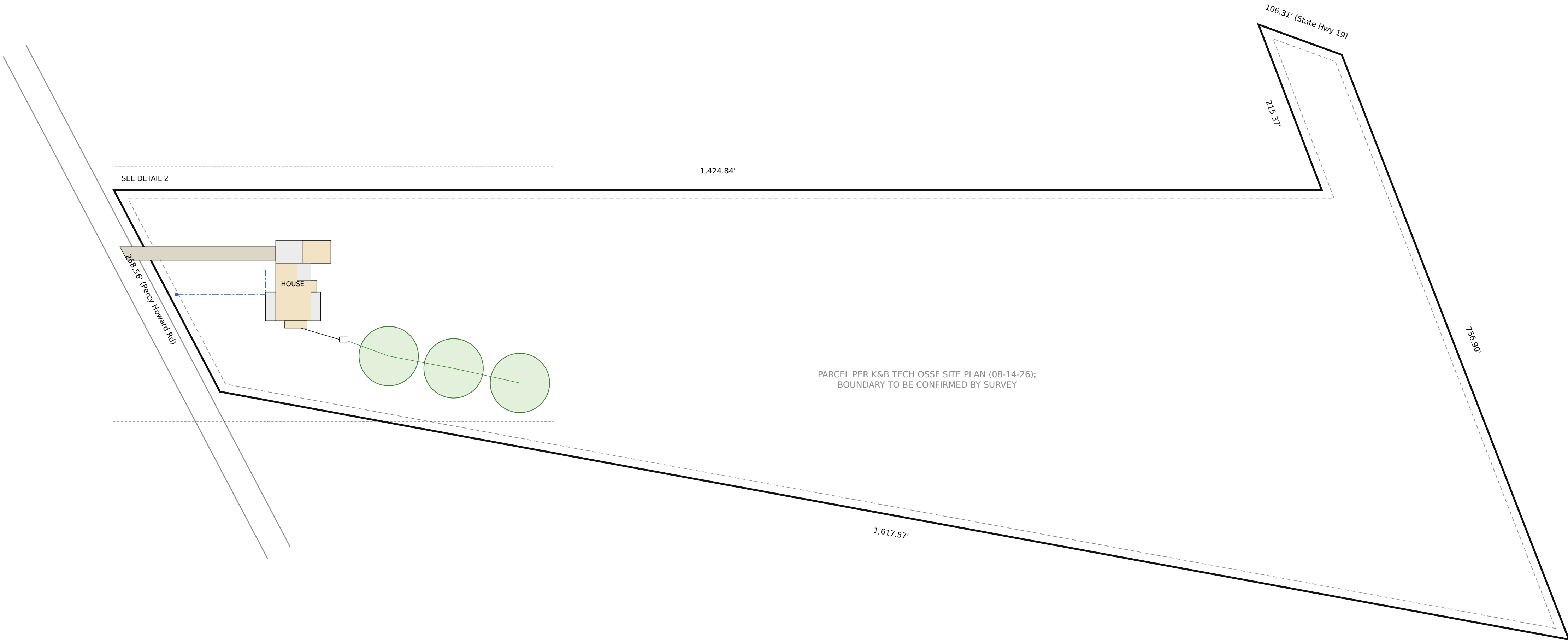
SITE PLAN — OVERALL PARCEL

SCALE: 1" = 80'-0"

2

SITE PLAN — HOUSE AREA

SCALE: 1" = 30'-0"



SITE NOTES

- Property line lengths, frontage and spray areas are taken from the K&B Tech OSSF site plan (sealed by J. Katambani, P.S. #3710, 08-14-26). Boundary, easements and building location to be confirmed by a registered surveyor.
- House location is approximate and follows the septic plan; the front faces Percy Howard Rd (north). Dimensions marked = are scaled and must be confirmed by survey stake-out before the county development permit.
- 10' setbacks shown dashed. Building must stay outside all setbacks and easements.
- Septic: aerobic Pro-Flo 800 PTSL, 1,000 gal pump tank at the west end, three 36" R spray circles. Keep spray areas at least 25'-30' from any slab (expansive clay); orchard, garden and pond clear of spray areas.
- Water: public meter and line from Percy Howard Rd. Electric service route per utility; well option: note 13.
- Driveway and culvert per Walker County Road & Bridge (county road).
- Flood zone: confirm on the FEMA map; finished floor at least 1' above BFE or 18" above grade.
- Grade falls 5% for 10' away from the house; no trees closer than their mature height; 30' cleared zone recommended.
- Address: settle one 911 address for all permits (septic plan shows 77340; geotech shows 17 Percy Howards Rd, 77320).
- Soils per the PA-14 site evaluation (08-14-26, App. 2026-0328, J. Katambani, OS-17329): two 64" borings, loam over clay loam over clay (Class II to IV); no water table or restrictive horizon; outside the 100-yr floodplain; slope 2-15%. K&B design: 480 gpd (300 + 180), spray rate 0.041 gal/sf/day, 11,707 sf required, 12,208 sf designed.
- Tank and spray circles are redrawn from the revised K&B site plan (Pro-Flo 800 PTSL, 1,000 g pump tank, two-way cleanouts), good to about 5'. K&B labels the circles 35' R (about 11,545 sf); they are drawn at 36' R so the three total about 12,214 sf, matching the 12,208 sf designed. K&B to confirm.
- Propane (proposed): 500 gal tank, 10' min. from buildings (25' if 1,000 gal). Buried 3/4" PE line, 18" deep, to a riser at the suite east wall; piping on P-101.
- Water well (proposed): about 5' east and 90' south of the carport corner; 100' min. from spray areas, 50' from the tank and sewer. Public meter is the alternative; confirm.

PRELIMINARY
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CONSTRUCTION

PEREZ RESIDENCE

Perez Ranch
43 Percy Howard Rd, Huntsville, TX
Walker County (unincorporated)
Owner: Carlos Perez

ENGINEER OF RECORD (TEXAS PE)

Seal, signature
and date

Name / firm / F-number:

REVISIONS

No.	Date	Description
0	10/05/2026	Preliminary for PE review
1	10/08/2026	Address; fireplace removed; vaults; BOM file renamed
2	10/08/2026	Septic per K&B plan
3	10/08/2026	Propane line; well

PREPARED FROM

Owner's 3D frame model and BOM
(Perez Ranch Material BOM.xlsx).
Room layout: Quality First Professional
Design Services plan set (rev. 03-25-26).
Geotech: SoilsTEC KT26G035B.
Septic: K&B Tech OSSF plan, 08-14-26.

SHEET TITLE

SITE PLAN
SCALE: AS NOTED

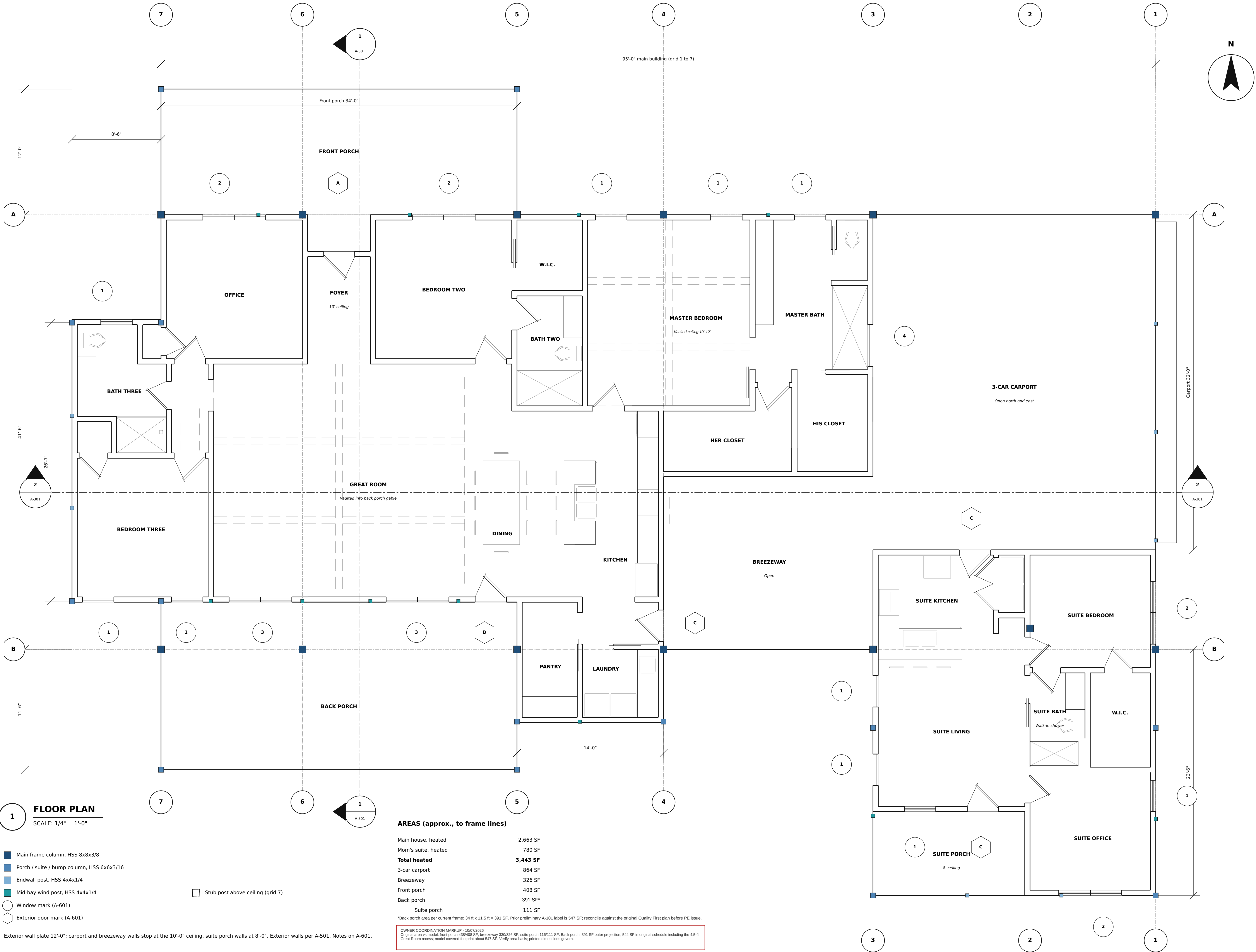
DATE
10/05/2026

DRAWN
Owner / Claude

SHEET

C-101

Preliminary design. The Texas PE's sealed drawings and calculations govern.



1 FLOOR PLAN
SCALE: 1/4" = 1'-0"

- Main frame column, HSS 8x8x3/8
- Porch / suite / bump column, HSS 6x6x3/16
- Endwall post, HSS 4x4x1/4
- Mid-bay wind post, HSS 4x4x1/4
- Window mark (A-601)
- Exterior door mark (A-601)
- Stub post above ceiling (grid 7)

Exterior wall plate 12'-0"; carport and breezeway walls stop at the 10'-0" ceiling, suite porch walls at 8'-0". Exterior walls per A-501. Notes on A-601.

AREAS (approx., to frame lines)

Main house, heated	2,663 SF
Mom's suite, heated	780 SF
Total heated	3,443 SF
3-car carport	864 SF
Breezeway	326 SF
Front porch	408 SF
Back porch	391 SF*
Suite porch	111 SF

*Back porch area per current frame: 34 ft x 11.5 ft = 391 SF. Prior preliminary A-101 label is 547 SF; reconcile against the original Quality First plan before PE issue.

OWNER COORDINATION MARKUP - 10/07/2026
Original area vs model: front porch 439/408 SF; breezeway 330/326 SF; suite porch 116/111 SF. Back porch: 391 SF outer projection, 544 SF in original schedule including the 4.5-ft Great Room recess; model covered footprint about 547 SF. Verify area basis; printed dimensions govern.

**PRELIMINARY
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PEREZ RESIDENCE

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43 Percy Howard Rd, Huntsville, TX
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ENGINEER OF RECORD (TEXAS PE)

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PREPARED FROM

Owner's 3D frame model and BOM
(Perez Ranch Material BOM.xlsx).
Room layout: Quality First Professional
Design Services plan set (rev. 03-25-26).
Geotech: SolisTEC KT26G035B.
Septic: K&B Tech OSSF plan, 08-14-26.

SHEET TITLE

FLOOR PLAN

SCALE: 1/4" = 1'-0"

DATE	DRAWN
10/05/2026	Owner / Claude

SHEET

A-101

Preliminary design. The Texas PE's sealed
drawings and calculations govern.

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PEREZ RESIDENCE

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Walker County (unincorporated)

Owner: Carlos Perez

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Seal, signature
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Geotech: SolisTEC KT26G035B.

Septic: K&B Tech OSSF plan, 08-14-26.

SHEET TITLE

EXTERIOR ELEVATIONS
NORTH AND SOUTH

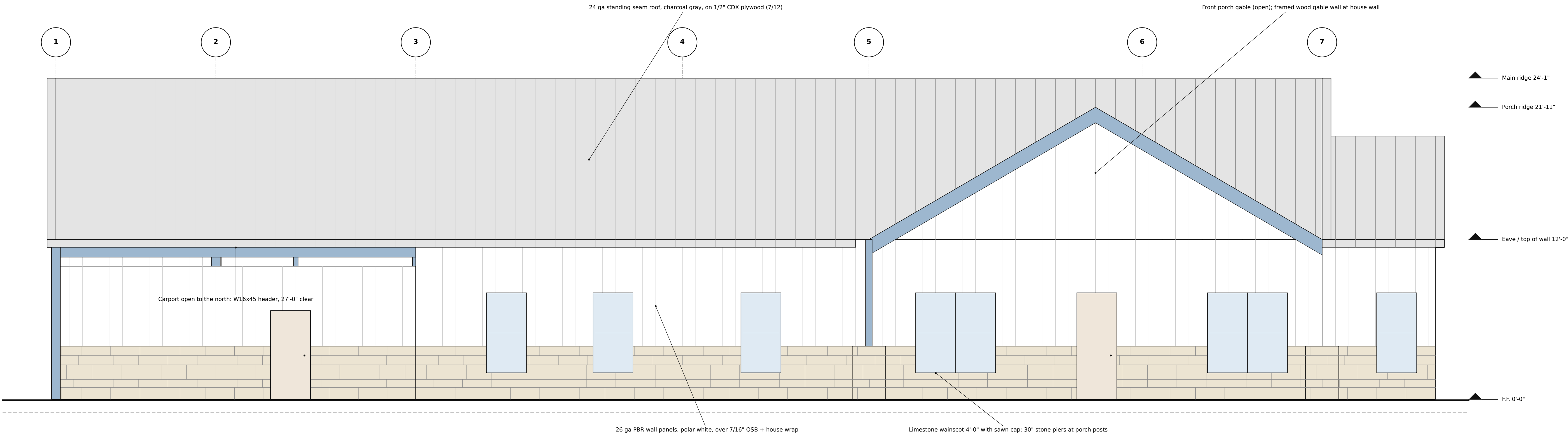
SCALE: 1/4" = 1'-0"

DATE	DRAWN
10/05/2026	Owner / Claude

SHEET

A-201

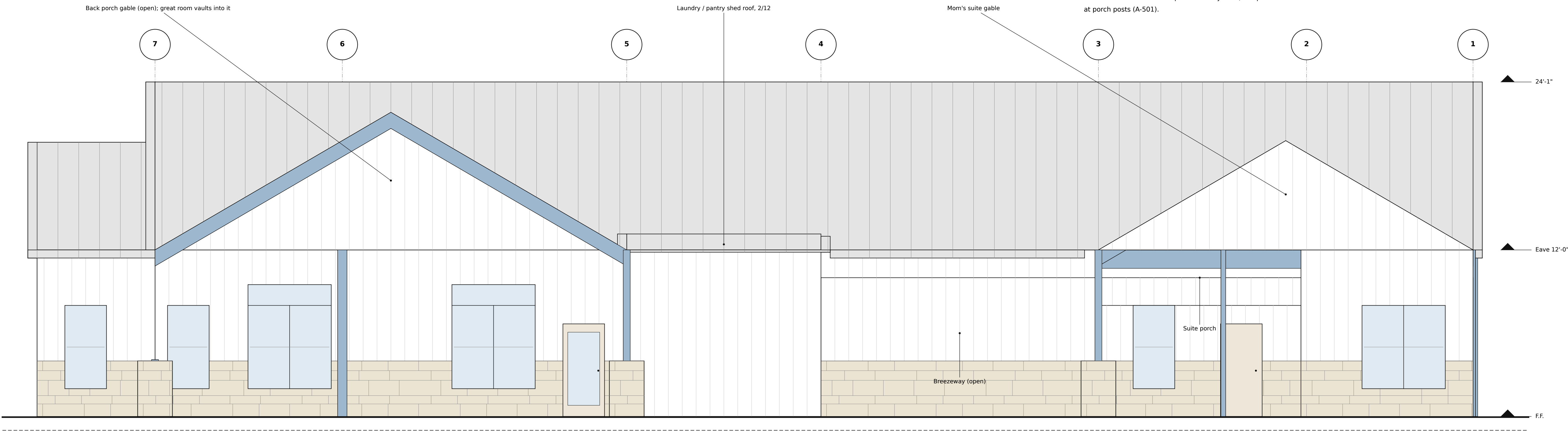
Preliminary design. The Texas PE's sealed
drawings and calculations govern.



1 NORTH ELEVATION (FRONT)
SCALE: 1/4" = 1'-0"

ELEVATION NOTES

1. Finished floor and pad height per survey and the PE's foundation design (2'-4" of select fill).
2. Windows and doors owner-supplied, DP-50 minimum; impact-rated recommended. See A-601.
3. Roof overhangs 12" at eaves, 8" at rakes; framing and clips per PE for overhang uplift.
4. Gutters 6", oversized at valleys; downspouts discharge 10'+ from the slab.
5. Limestone: 4' wainscot except the laundry shed; 30" piers at porch posts (A-501).



2 SOUTH ELEVATION (REAR)
SCALE: 1/4" = 1'-0"

PRELIMINARY
FOR PE REVIEW

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43 Percy Howard Rd, Huntsville, TX
Walker County (unincorporated)

Owner: Carlos Perez

ENGINEER OF RECORD (TEXAS PE)

Seal, signature
and date

Name / firm / F-number:

REVISIONS		
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PREPARED FROM

Owner's 3D frame model and BOM
(Perez Ranch Material BOM.xlsx).

Room layout: Quality First Professional
Design Services plan set (rev. 03-25-26).

Geotech: SolisTEC KT26G035B.

Septic: K&B Tech OSSF plan, 08-14-26.

SHEET TITLE

EXTERIOR ELEVATIONS
EAST AND WEST

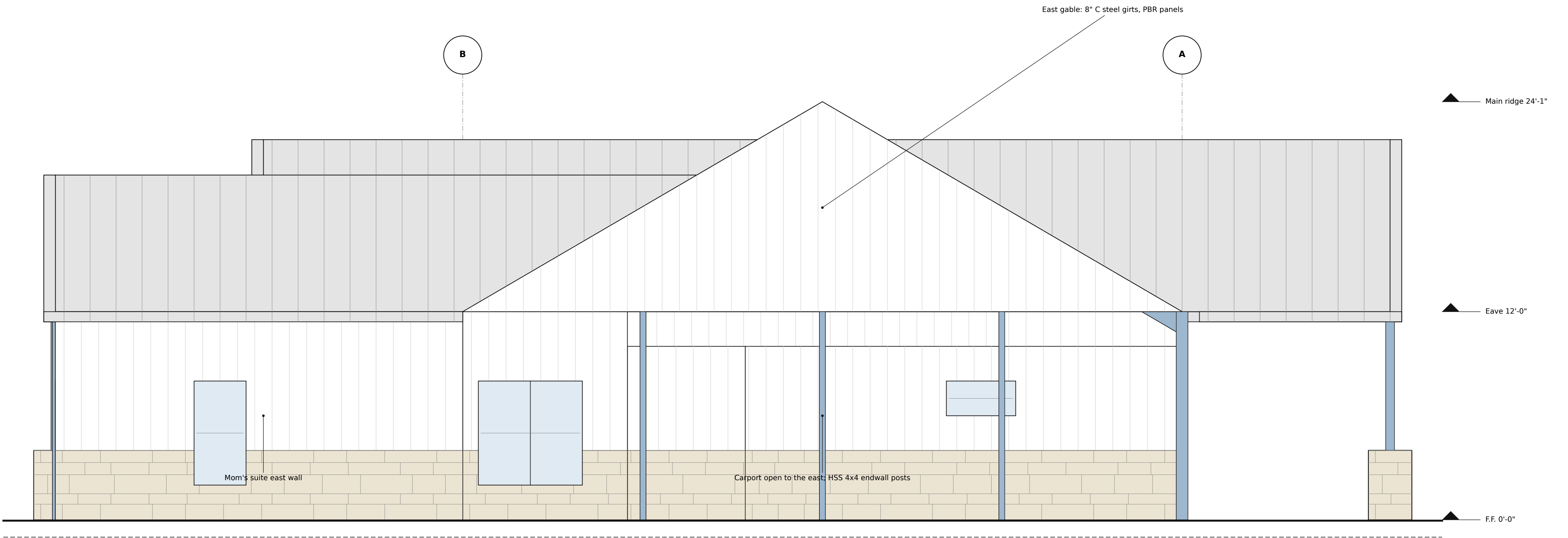
SCALE: 1/4" = 1'-0"

DATE	DRAWN
10/05/2026	Owner / Claude

SHEET

A-202

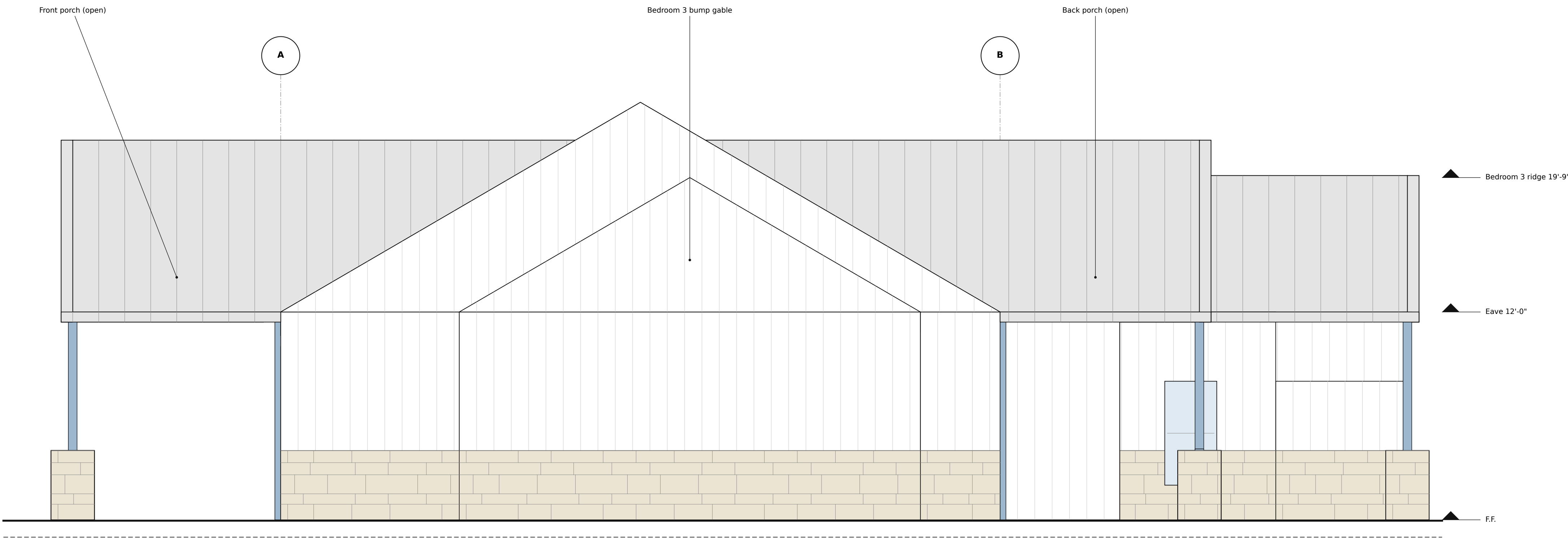
Preliminary design. The Texas PE's sealed drawings and calculations govern.



ELEVATION NOTES

- Gable triangles above the eave at grids 1 and 7, the suite rear gable and the bedroom 3 gable are framed with 8" C 14 ga girts (G90) at 4' o.c. Walls below the eave are 2x6 wood.
- Porch gables B and C are open at the outer end; the gable at the house wall is a framed wood wall with PBR panels (no glass).
- Exposed steel at the carport, porches and header is hot-dip galvanized (ASTM A123) or zinc-rich primer + epoxy + urethane.
- Lightning protection (NFPA 780) bonded to the steel frame; down conductors at the building corners.

1 EAST ELEVATION (CARPORT SIDE)
SCALE: 1/4" = 1'-0"



2 WEST ELEVATION
SCALE: 1/4" = 1'-0"

PRELIMINARY
FOR PE REVIEW

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CONSTRUCTION

PEREZ RESIDENCE

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43 Percy Howard Rd, Huntsville, TX
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Owner: Carlos Perez

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Seal, signature
and date

Name / firm / F-number:

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PREPARED FROM

Owner's 3D frame model and BOM
(Perez Ranch Material BOM.xlsx).

Room layout: Quality First Professional
Design Services plan set (rev. 03-25-26).

Geotech: SolisTEC KT26G035B.

Septic: K&B Tech OSSF plan, 08-14-26.

SHEET TITLE

BUILDING SECTIONS

SCALE: 1/4" = 1'-0"

DATE	DRAWN
10/05/2026	Owner / Claude

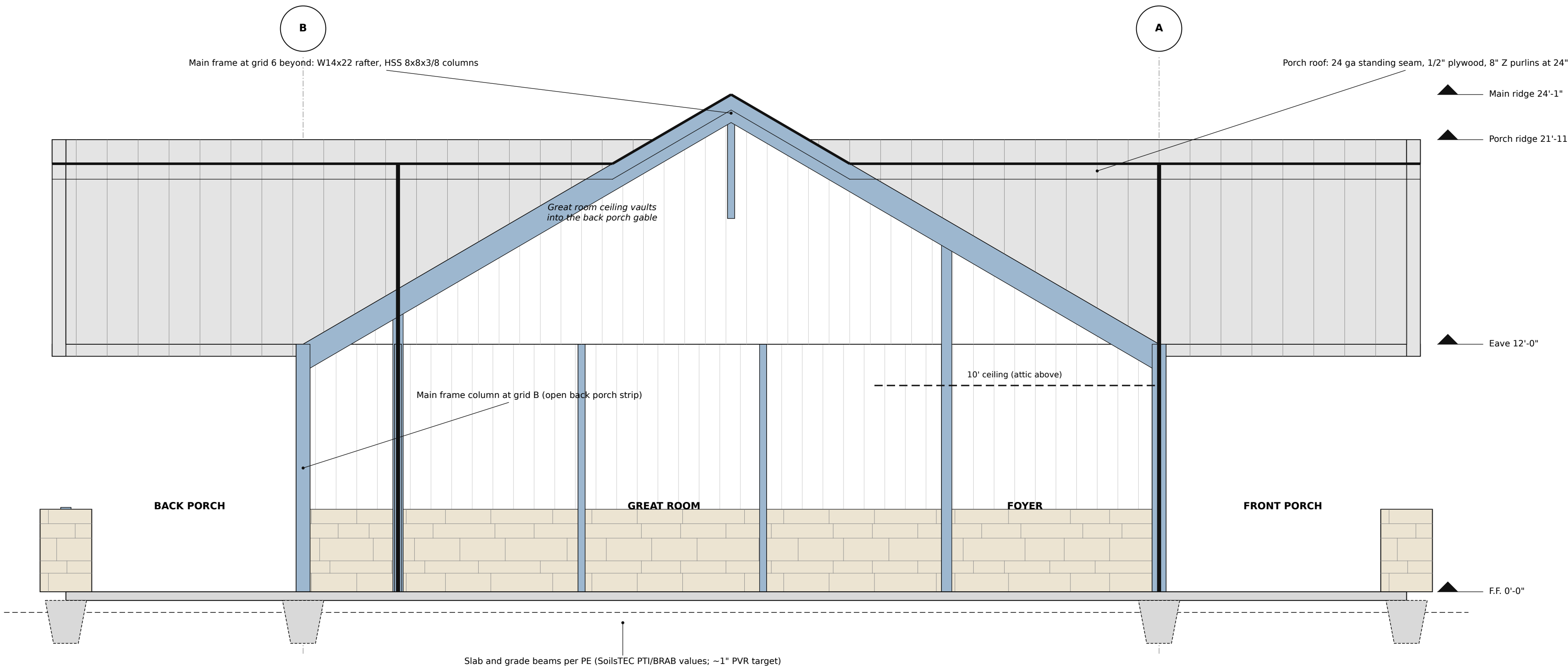
SHEET

A-301

Preliminary design. The Texas PE's sealed
drawings and calculations govern.

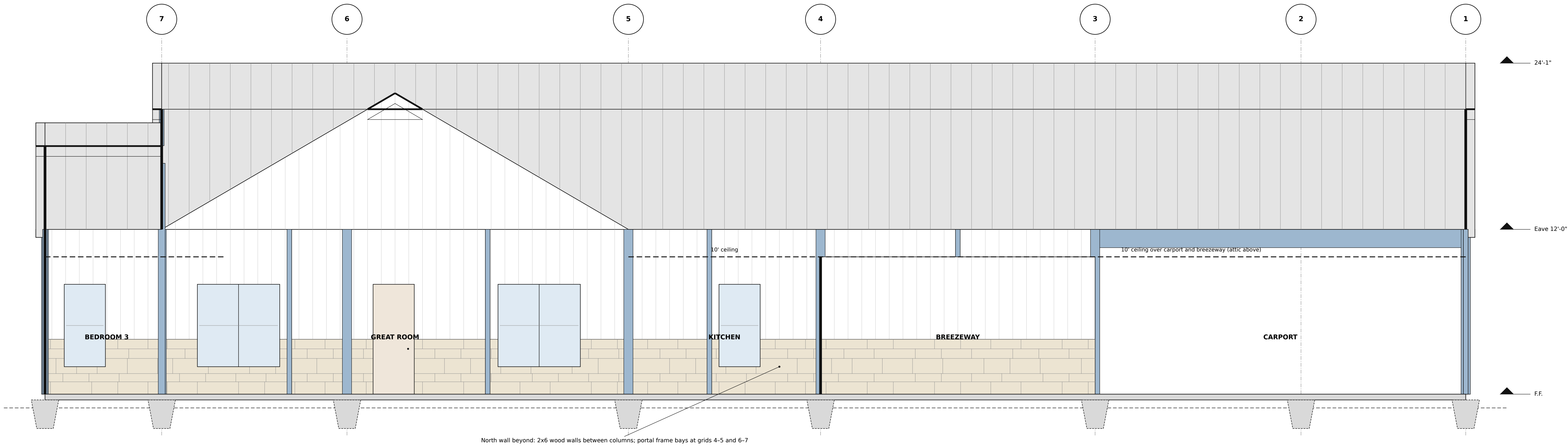
SECTION NOTES

- Sections are cut through the exterior shell; interior partitions are not shown.
- Main roof is removed under the porch gables. The porch roofs run to valleys on the main roof. The great room vaults up into the back porch gable (PE to detail valleys, cut purlins and the grid 6 frame through the vault); the front porch gable is attic space over the flat 10' ceilings of the foyer, office and bedroom 2.
- Ceilings: 10'-0" flat, except the great room vault (10'-0" to 18'-0" at the ridge, Grid 6) and the master bedroom vault (10'-0" to 12'-0" at the ridge); Mom's suite is 8'-0" flat. Attic space above the ceilings is shown in the 3D model (study only).
- Roof assembly: 24 ga standing seam (UL 580 Class 90 / FM 4471 clips) on synthetic underlayment, 1/2" CDX plywood, 8" Z 14 ga purlins at 24" o.c.; closed-cell spray foam under the deck.
- Wall assembly: see A-501.
- Finished floor at least 1' above BFE or 18" above grade; pad per geotech.



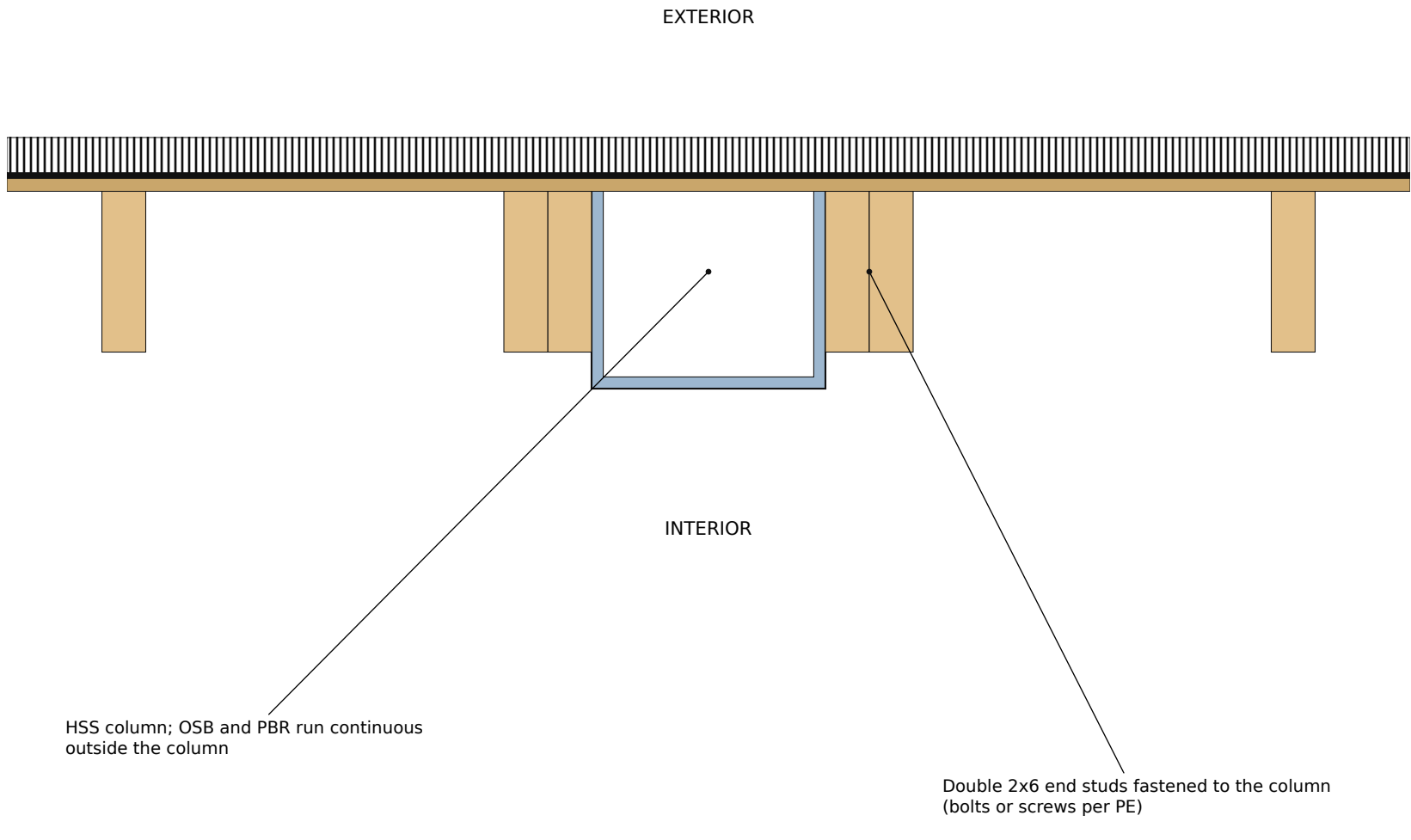
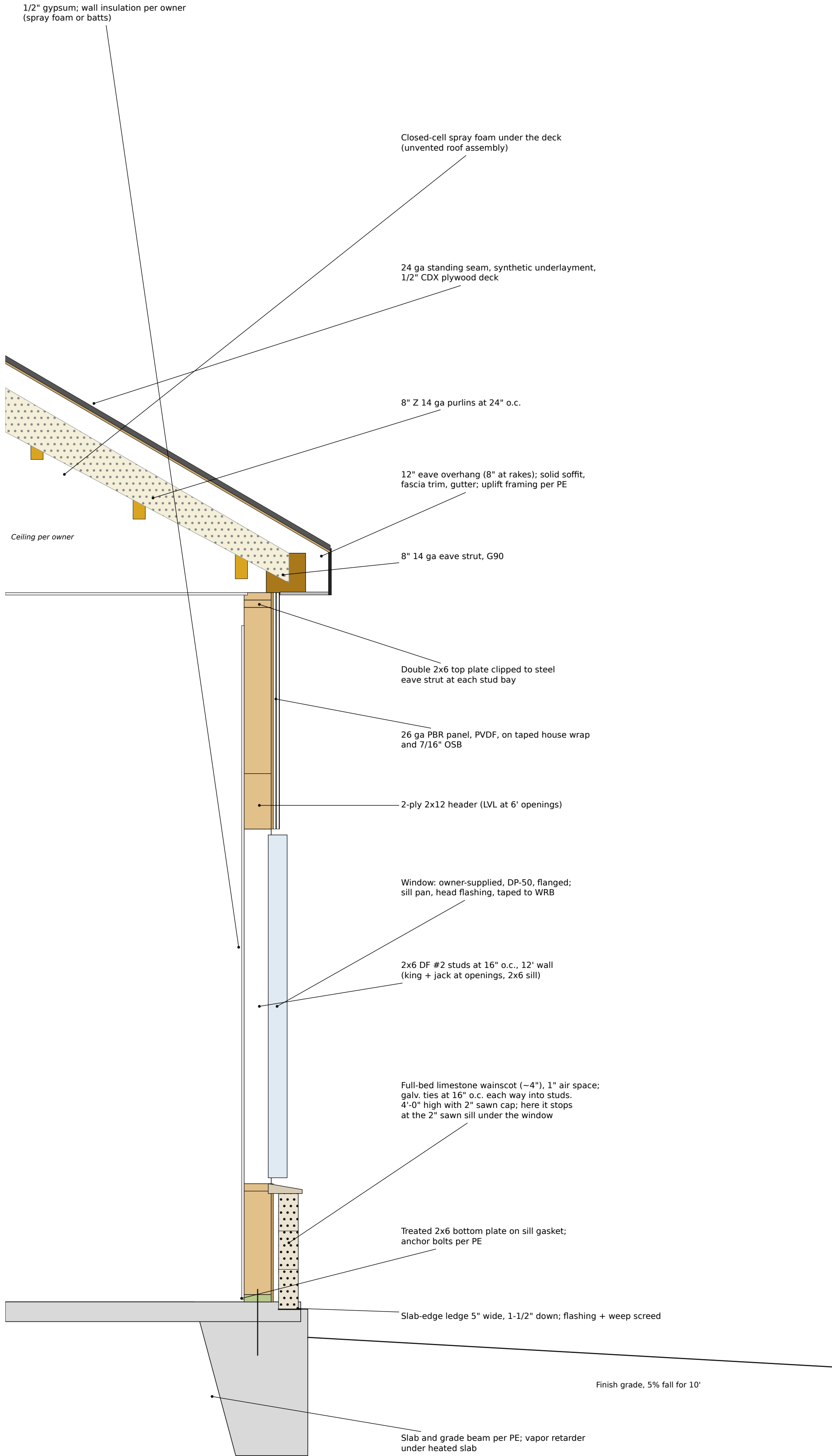
1 SECTION 1 — FOYER AND GREAT ROOM

SCALE: 1/4" = 1'-0"



2 SECTION 2 — LONGITUDINAL

SCALE: 1/4" = 1'-0"



WALL NOTES

- Wood: 2x6 Douglas fir #2 (SPF or SYP require 12" o.c.); treated bottom plate; double top plate. Stud check 0.72, deflection L/272.
- Sheathing: 7/16" OSB nailed per PE (wall shear where required). Taped house wrap lapped shingle-style over flashing.
- Openings: flanged windows, sill pan, jamb and head flashing; DP-50 minimum. Exterior doors: DP-50 with sill pan.
- Wall panels: 26 ga PBR with PVDF finish over 7/16" OSB, from the stone cap to the eave; panel maker's fasteners at the maker's tested spacing for 42 psf ASD (corner zones). Z-flashing over the stone cap, lapped under the WRB.
- Limestone wainscot: full-bed chopped Texas limestone (~4", ~45 psf), 4'-0" high on all house walls except the laundry/pantry shed, on a 5" slab-edge ledge 1-1/2" below the floor. 1" air space, corrugated galvanized ties at 16" o.c. each way into studs (TMS 402 high-wind spacing), weep screed and flashing at the base, weeps at 33" o.c., 2" sawn cap and window sills sloped out.
- Keep plumbing supply lines out of exterior walls; insulate.
- Seal all penetrations; closure strip and flashing where the PBR meets the stone cap.

OWNER COORDINATION MARKUP - 10/07/2026
Omit 4-ft limestone wainscot at the carport/master bath wall, carport/suite wall, master-closet/foreneway wall, and kitchen/foreneway wall. Continue metal cladding to slab; retain framing. Concrete slab-edge ledge/reinforcement remains subject to PE review. BOM: wainscot 864 SF, wall panels 4,542 9 SF. NOT FOR CONSTRUCTION.

2 PLAN DETAIL: WOOD WALL AT STEEL COLUMN

SCALE: 1-1/2" = 1'-0" (approx.)

PRELIMINARY
FOR PE REVIEW
NOT FOR
CONSTRUCTION

PEREZ RESIDENCE

Perez Ranch
43 Percy Howard Rd, Huntsville, TX
Walker County (unincorporated)

Owner: Carlos Perez

ENGINEER OF RECORD (TEXAS PE)



Name / firm / F-number:

REVISIONS

No.	Date	Description
0	10/05/2026	Preliminary for PE review
1	10/08/2026	Address; fireplace removed; vaults; BOM file renamed

PREPARED FROM

Owner's 3D frame model and BOM
(Perez Ranch Material BOM.xlsx).
Room layout: Quality First Professional
Design Services plan set (rev. 03-25-26).
Geotech: SolisTEC KT26G035B.
Septic: K&B Tech OSSF plan, 08-14-26.

SHEET TITLE

TYPICAL WALL SECTION AND WALL DETAILS

AS NOTED

DATE DRAWN
10/05/2026 Owner / Claude

SHEET

A-501

Preliminary design. The Texas PE's sealed drawings and calculations govern.

1 TYPICAL EXTERIOR WALL SECTION

SCALE: 3/4" = 1'-0"

Exterior Window Schedule

Mark	Qty	Size (W × H)	Type	Head	Sill	Locations
1	10	3'-0" × 6'-0"	Single, vinyl clad insulated	8'-0"	2'-0"	Bath 3, Bedroom 3 (rear), Master bath, Master bedroom, Suite living, Suite living (porch), Suite office
2	4	(2) 3'-0" × 6'-0"	Mulled pair, vinyl clad insulated	8'-0"	2'-0"	Bedroom 2, Office, Suite bedroom, Suite bedroom (rear)
3	2	(2) 3'-0" × 6'-0" + transom	Mulled pair with transom	9'-6"	2'-0"	Great room (rear)
4	1	4'-0" × 2'-0"	Awning/fixd, vinyl clad insulated	8'-0"	6'-0"	Master shower (carport side)

Exterior Door Schedule

Mark	Qty	Size (W × H)	Type	Locations
A	1	3'-0" × 8'-0"	Exterior arched entry door	Foyer - front entry
B	1	3'-0" × 6'-8"	Exterior French door	Dining to back porch
C	3	3'-0" × 6'-8"	Exterior panel door	Kitchen hall to breezeway, Suite kitchen to carport, Suite porch door

Totals: 17 window units, 5 exterior doors. All interior passage doors 3'-0" (36") wide for wheelchair access; rough openings 3'-2".

Window and Door Requirements

- All windows and exterior doors are owner-supplied and must carry a design pressure rating of DP-50 or better (corner-zone demand about 42 psf ASD at 140 mph, partially enclosed).
- Impact-rated units (ASTM E1886 / E1996) are recommended; they keep the envelope closed in debris-laden wind.
- Exterior doors: DP-50 rated assemblies with multi-point latching where available; sill pans at every door.
- Install per the manufacturer's tested installation instructions (fastener type and spacing into the 2x6 framing) so the DP rating holds.
- Rough openings: 2x6 king and jack studs each side, 2-ply 2x12 header (LVL at 6' openings), 2x6 sill.
- Egress: bedroom windows to meet IRC R310 (net clear opening, sill height ≤ 44").
- Safety glazing where required by IRC R308 (doors, near tubs/showers, low sills).

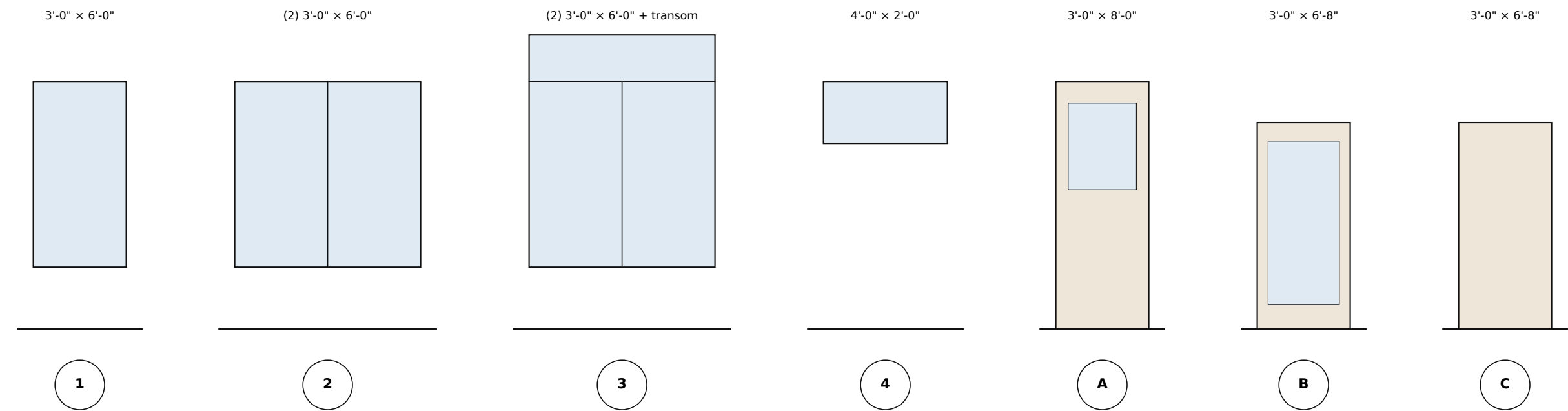
Mom's Suite: Aging in Place

- Single level throughout. All interior passage doors 36" wide (rough opening 3'-2") for a walker or wheelchair.
- Zero-step entries: suite entry doors (kitchen to carport, porch door) with flush or 1/2" max beveled thresholds; flush floor transitions between rooms.
- Bath: curbless walk-in shower (slab recessed per PE, linear drain), built-in or fold-down shower bench, handheld shower on a slide bar, comfort-height toilet (17"-19"), 60" clear turning circle, lever faucets.
- Grab-bar blocking: solid 2x wood blocking in the bath walls at 30"-40" AFF around the toilet, shower and bench, so grab bars can be added at any time.
- Hardware and electrical: lever door handles; rocker switches at 42"-44" AFF; receptacles at 18" AFF; night lights in the bath and hall.
- Alarms: smoke and CO alarms in the suite interconnected with the main house; call button or intercom from the suite bedroom and bath to the main house.
- Suite A/C: return-air filter grille reachable from the floor (no filter change at the attic air handler); thermostat at 48" AFF max.

General Notes

- These drawings show the current agreed design: steel rigid frames with 2x6 wood exterior walls, 12'-0" walls, 7/12 roofs, great room vaulted into the back porch gable, open carport at the east end.
- Design basis: ASCE 7-22, 140 mph ultimate, Exposure C, Risk Category II, whole house partially enclosed (GCpi ±0.55).
- Code: Texas Local Government Code §233.153 (IRC as applied in Huntsville, or the IRC as of May 1, 2008) with three inspections under §233.154.
- Room layout, interior partitions, fixtures, cabinets, plumbing and electrical follow the Quality First plan set (rev. 03-25-26). Where that set differs, these drawings govern: 10'-0" walls (vaulted great room and master ceilings per note 9), walk-in showers (no tubs), no fireplace or chimney, no gable glass, no front gable at the master bedroom.
- Dimensions are to grid lines and column centerlines unless noted. Do not scale the drawings; verify in the field.
- Finished floor elevation, pad height and building location per survey, flood-zone check and the PE's foundation design.
- Septic (OSSF) per K&B Tech plan; spray circles to be relocated 25'-30'+ from any slab before permitting.
- All showers are walk-in. All plumbing supply lines in interior walls.
- Interior walls 10'-0" tall (ceilings at 10'), except the great room, which has a vaulted ceiling (10'-0" at the walls rising to 18'-0" at the ridge, Grid 6); the master bedroom, which has a vaulted ceiling (10'-0" rising to 12'-0" at the ridge); and Mom's suite, which has 8'-0" walls and ceilings (her porch also has an 8'-0" ceiling). All interior passage doors 36" wide.

Unit Elevations (exterior view, 1/4" = 1'-0")



PRELIMINARY

FOR PE REVIEW

NOT FOR

CONSTRUCTION

Perez Residence

Perez Ranch
43 Percy Howard Rd, Huntsville, TX
Walker County (unincorporated)

Owner: Carlos Perez

Engineer of Record (Texas PE)



Name / firm / F-number:

Revisions

No.	Date	Description
0	10/05/2026	Preliminary for PE review
1	10/08/2026	Address; fireplace removed; vaults; BOM file renamed

Prepared From

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Design Services plan set (rev. 03-25-26).
Geotech: SolisTEC KT26G035B.
Septic: K&B Tech OSSF plan, 08-14-26.

Sheet Title

Door and Window
Schedules, Notes

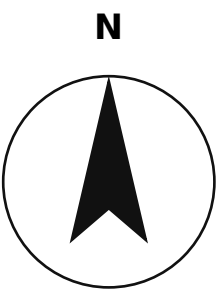
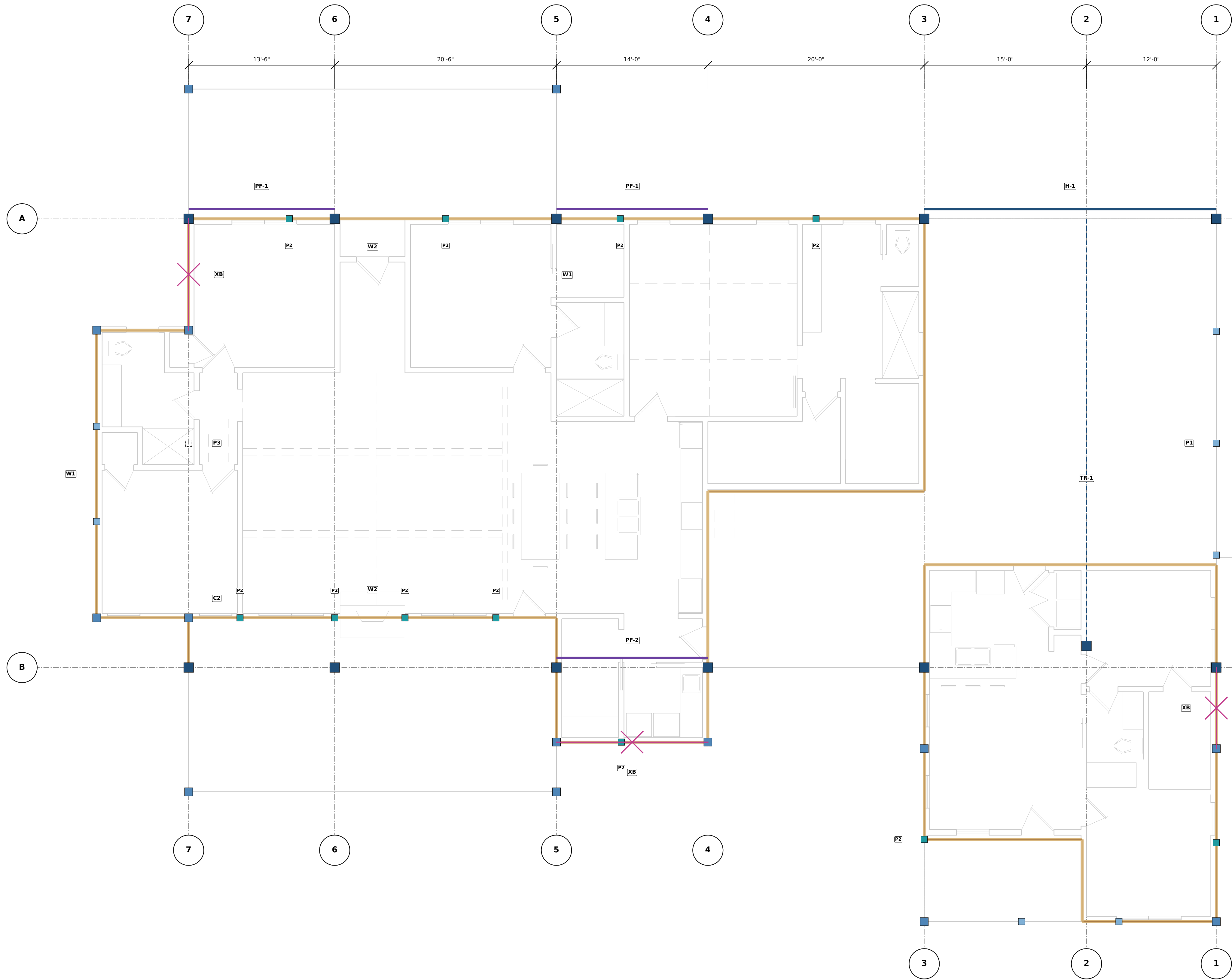
DATE	DRAWN
10/05/2026	Owner / Claude

Sheet

A-601

Preliminary design. The Texas PE's sealed drawings and calculations govern.





MEMBER SCHEDULE

Mark	Member	Size	Notes
C1	Main frame column	HSS 8x8x3/8	None at 2A; 2B is 2'-0" N of grid B: fur its wall to 9"
C2	Porch, suite, bump, shed columns	HSS 6x6x3/16	Grid 7 bump corners run to main rafter
P1	Endwall / carport posts	HSS 4x4x1/4	Grid 1, bump and suite gables
P2	Mid-bay wind posts (11)	HSS 4x4x1/4	Top-of-wall span ≤ 10'-5" on 12' walls
P3	Grid 7 gable stub post	HSS 4x4x1/4	Above bedroom 3 ceiling, on bump rafter
PF-1	Portal frame (north wall)	W12x26 beam, C1 legs	Bays 4-5 and 6-7; moment connections
PF-2	Portal frame (grid B, laundry bay)	W12x26 beam at 10', C1 legs	Kitchen-laundry opening stays clear
XB	Wall rod X-bracing	7/8" rod, A36, turnbuckle	Laundry rear wall, grid 7 north bay, suite east
H-1	Carport header	W16x45, 27'-0" clear	Carries grid 2; sized for a future attic floor
TR-1	Tie rod, grid 2 frame	1" rod with turnbuckle	At the eave; ~7.8 k tension
W1	Exterior wood wall	2x6 DF #2 at 16" o.c.	See A-501; treated bottom plate
W2	Porch gable wall at house	2x6 DF #2 at 16" o.c.	Framed above the eave, no glass

FRAMING NOTES

- Lateral system: rigid frames across the building; portal frames PF-1 on the north wall and PF-2 in the grid B laundry bay, rod X-bracing (XB) end to end; plywood roof deck as diaphragm per PE. Wind posts in PF-1 bays bolt to the portal beam (PE detail).
- Walls that stop at a ceiling (carport and breezeway walls at 10'; suite porch walls at 8') are braced at the top by the ceiling / attic floor diaphragm; 12' walls by the eave strut and P2 posts.
- Wood walls: 2x6 Douglas fir #2 at 16" o.c., treated bottom plate anchored to slab, double top plate clipped to the steel eave strut. 7/16" OSB, taped house wrap, 26 ga PBR above a 4'-0" limestone wainscot.
- Headers: 2-ply 2x12 (LVL at 6' openings); king + jack studs each side; 2x6 sills. Door and window schedule on A-601.
- Interior partitions: 2x4 SPF #2 at 16" o.c., non-bearing, treated bottom plate and double top plate; 10'-0" tall, except 12'-0" around the great room and 8'-0" in Mom's suite. Door rough openings 3'-2" for 36" doors (wheelchair access).
- Gable triangles above the eave at grids 1 and 7, the bump gable and the suite rear gable: 8" C 14 ga girts at 4' o.c., spans ≤ 10'-5".
- All structural steel A500 Gr C (HSS) / A992 (W shapes). Bolted moment connections preferred; field welds by AWS D1.1 certified welders.
- Cap and seal HSS columns. Exposed steel hot-dip galvanized or zinc-rich + epoxy + urethane.

1 WALL, COLUMN AND BRACING PLAN
SCALE: 3/16" = 1'-0"

- Wood stud wall (W1 / W2)
- Portal frame beam
- Rod X-bracing in wall plane
- Tie rod

PRELIMINARY
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PEREZ RESIDENCE

Perez Ranch
43 Percy Howard Rd, Huntsville, TX
Walker County (unincorporated)

Owner: Carlos Perez

ENGINEER OF RECORD (TEXAS PE)



Name / firm / F-number:

REVISIONS

No.	Date	Description
0	10/05/2026	Preliminary for PE review
1	10/08/2026	Address, fireplace removed; vaults, BOM file renamed

PREPARED FROM

Owner's 3D frame model and BOM
(Perez Ranch Material BOM.xlsx).
Room layout: Quality First Professional
Design Services plan set (rev. 03-25-26).
Geotech: SolisTEC KT26G035B.
Septic: K&B Tech OSSF plan, 08-14-26.

SHEET TITLE

WALL, COLUMN AND
BRACING PLAN

SCALE: 3/16" = 1'-0"

DATE
10/05/2026

DRAWN
Owner / Claude

SHEET

S-201

Preliminary design. The Texas PE's sealed drawings and calculations govern.

Preliminary design. The Texas PE's sealed drawings and calculations govern.

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PEREZ RESIDENCE

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43 Percy Howard Rd, Huntsville, TX
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Owner: Carlos Perez

ENGINEER OF RECORD (TEXAS PE)

Seal, signature
and date

Name / firm / F-number:

REVISIONS

No.	Date	Description
0	10/05/2026	Preliminary for PE review
1	10/08/2026	Address; fireplace removed; vaults; BOM file renamed
2	10/08/2026	Footings on piers

PREPARED FROM

Owner's 3D frame model and BOM
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Room layout: Quality First Professional
Design Services plan set (rev. 03-25-26).
Geotech: SolisTEC KT26G035B.
Septic: K&B Tech OSSF plan, 08-14-26.

SHEET TITLE

FRAME ELEVATIONS

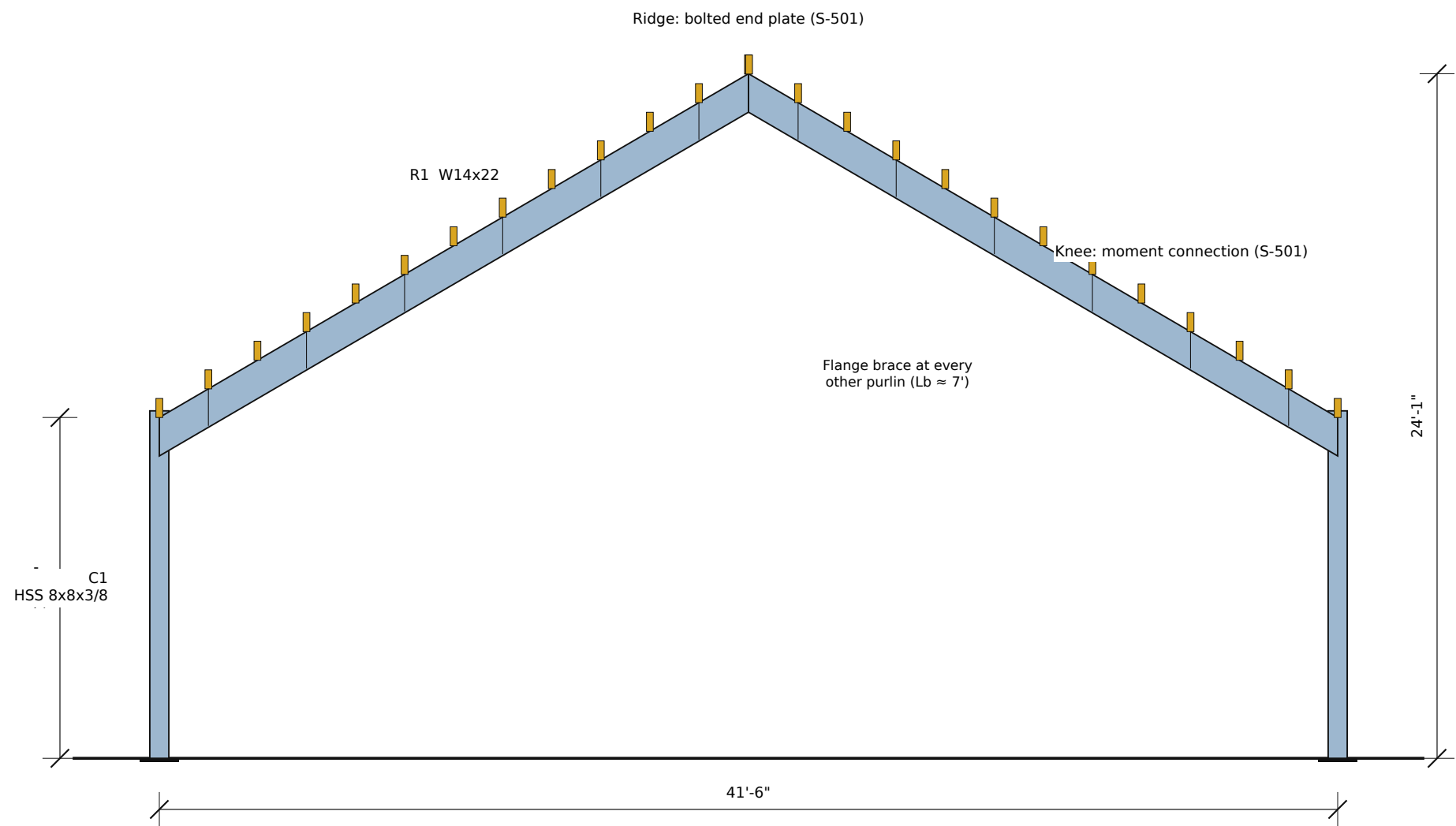
SCALE: 3/16" = 1'-0"

DATE	DRAWN
10/05/2026	Owner / Claude

SHEET

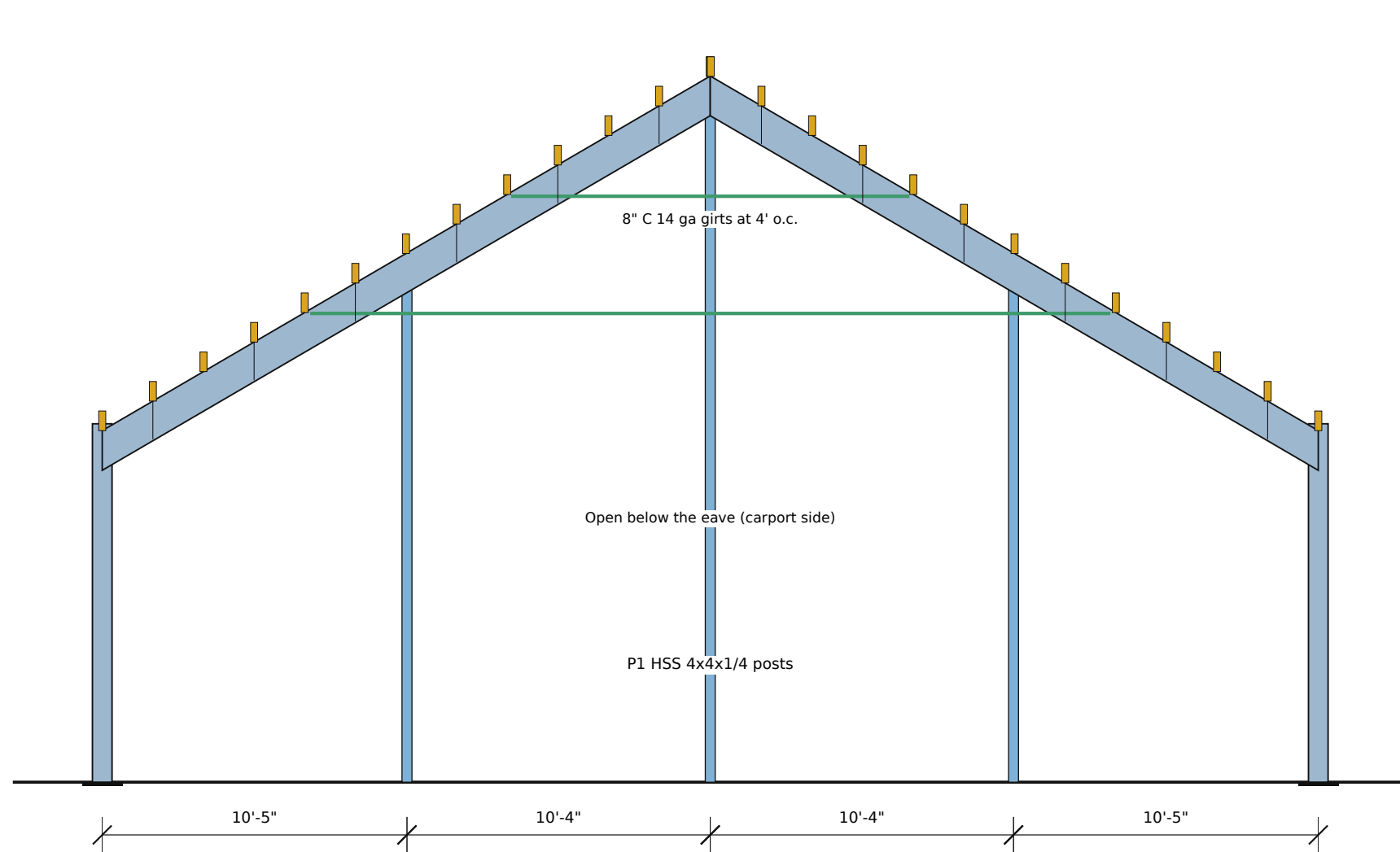
S-401

Preliminary design. The Texas PE's sealed
drawings and calculations govern.



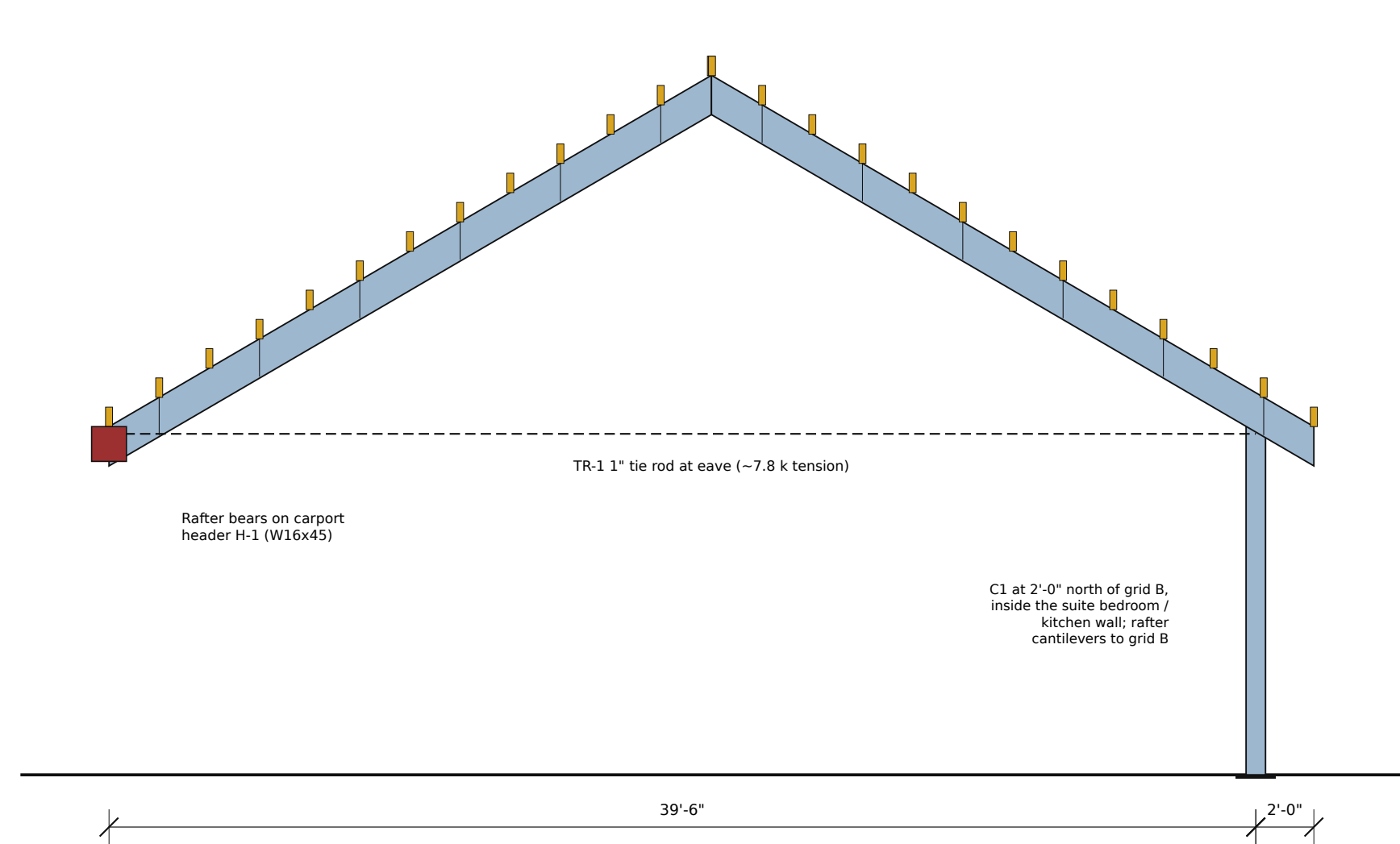
1 TYPICAL MAIN FRAME, GRIDS 3-6

SCALE: 3/16" = 1'-0"



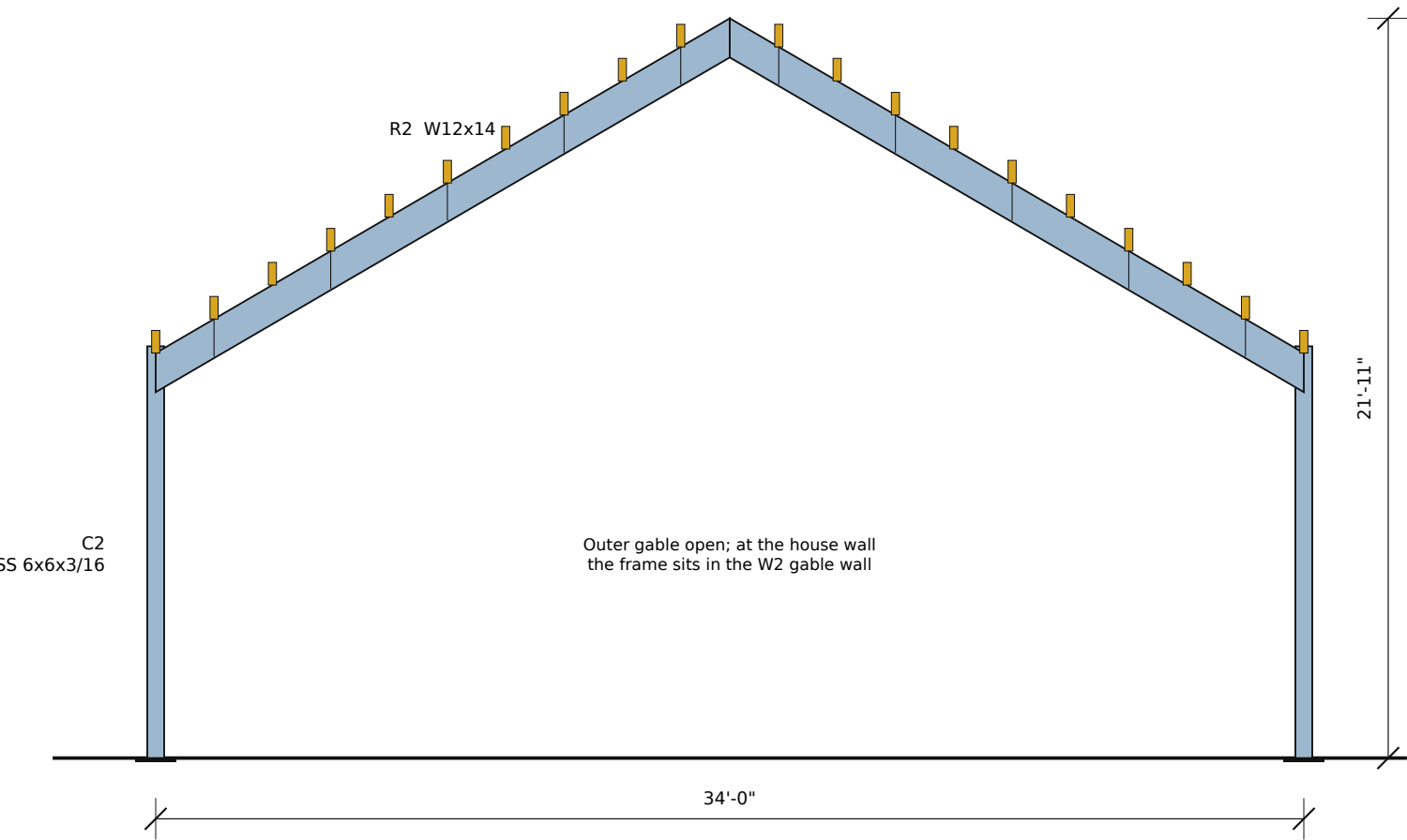
2 ENDWALL FRAME, GRID 1

SCALE: 3/16" = 1'-0"



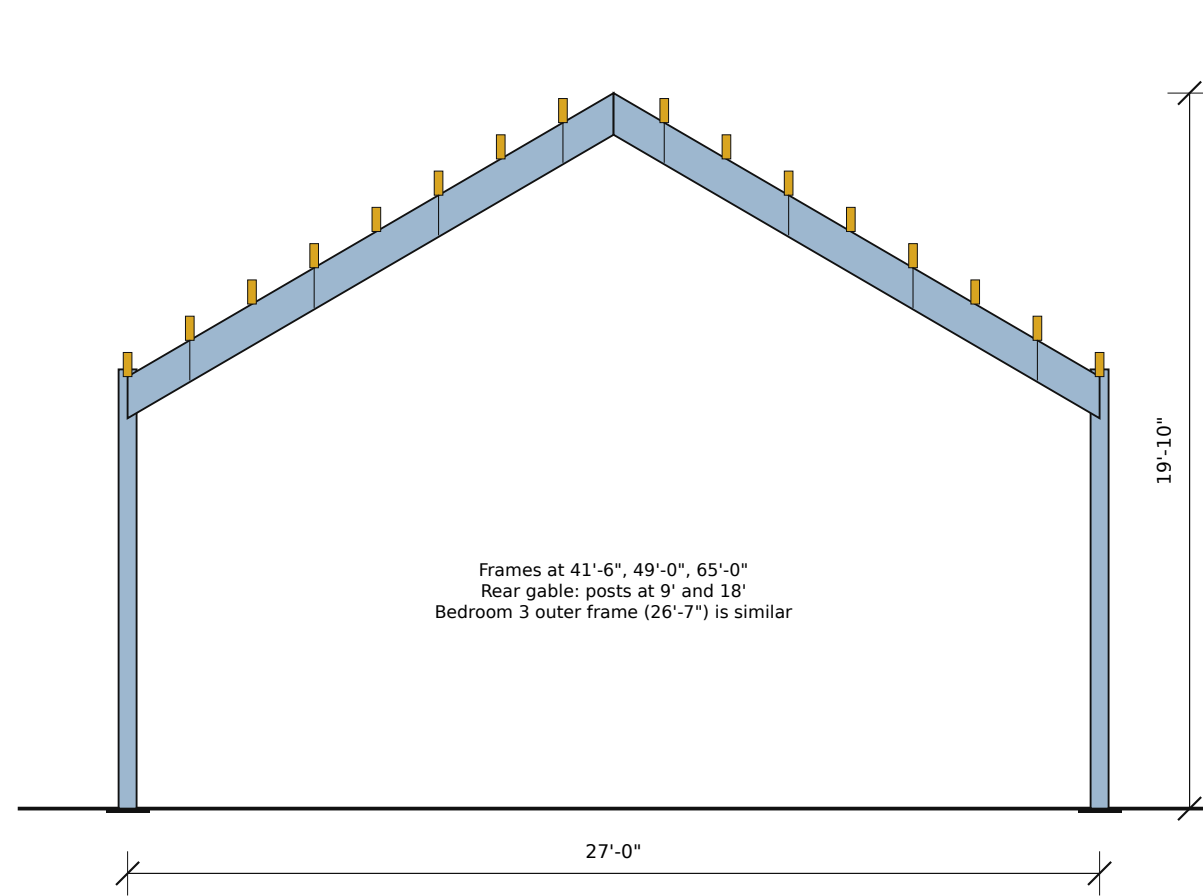
3 GRID 2 FRAME (CARPORT)

SCALE: 3/16" = 1'-0"



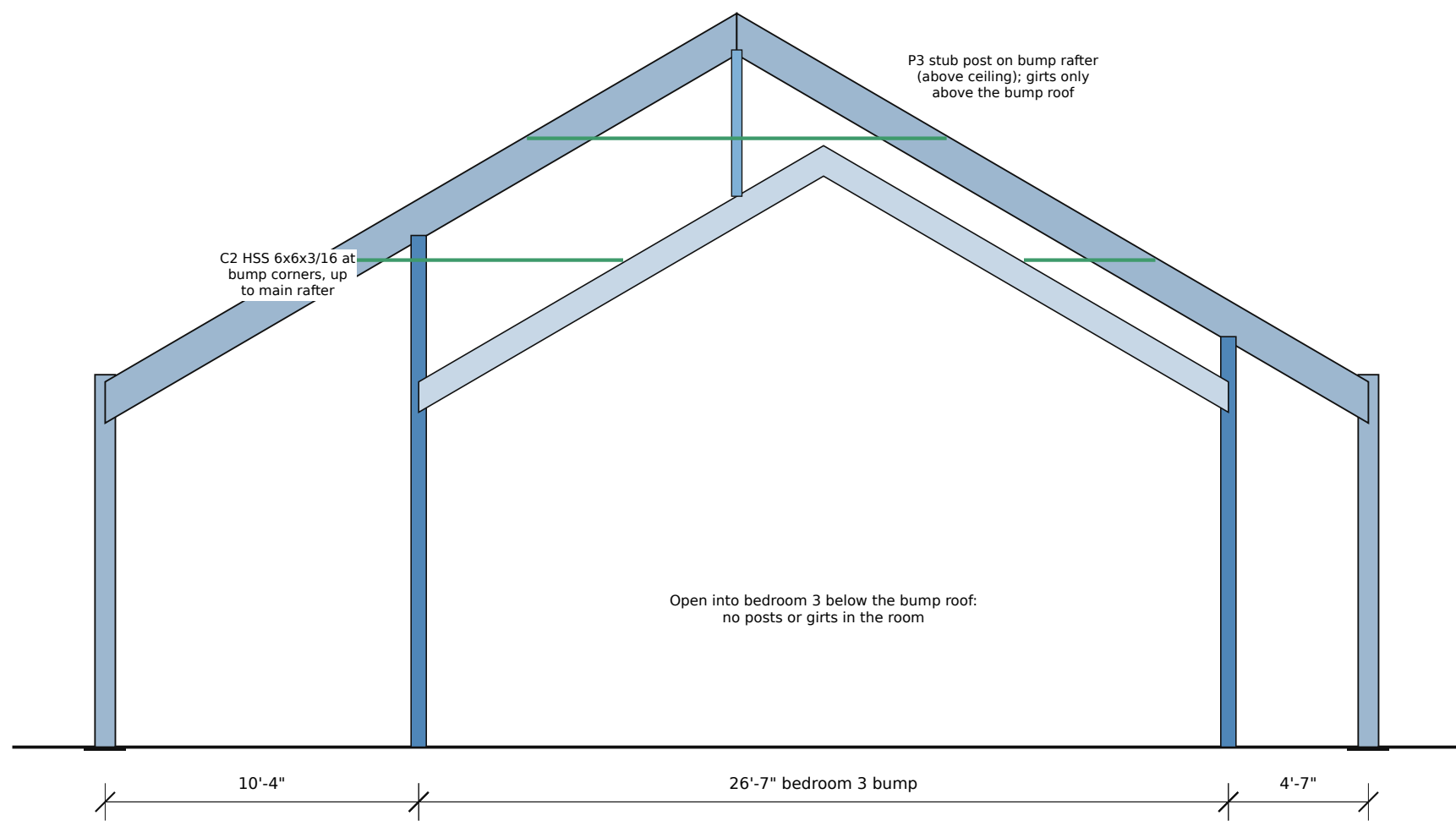
4 PORCH FRAMES B AND C

SCALE: 3/16" = 1'-0"



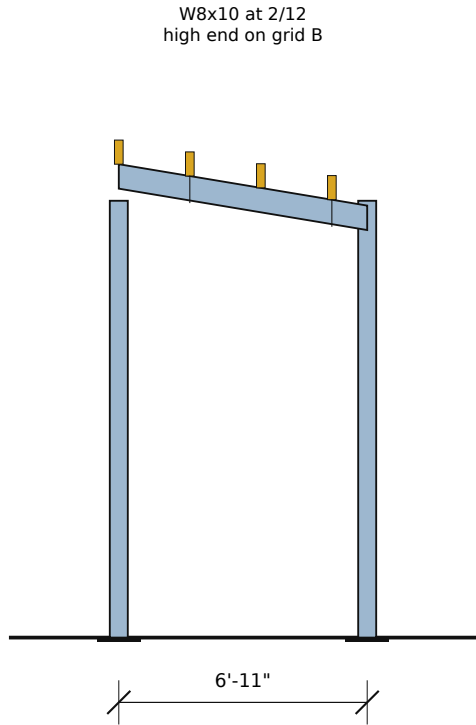
5 MOM'S SUITE FRAMES (E)

SCALE: 3/16" = 1'-0"



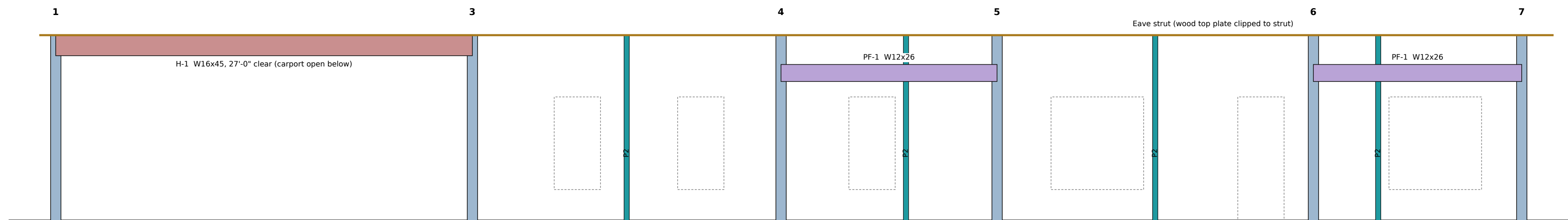
6 GRID 7 AT BEDROOM 3 BUMP

SCALE: 3/16" = 1'-0"



7 SHED FRAME (D)

SCALE: 3/16" = 1'-0"

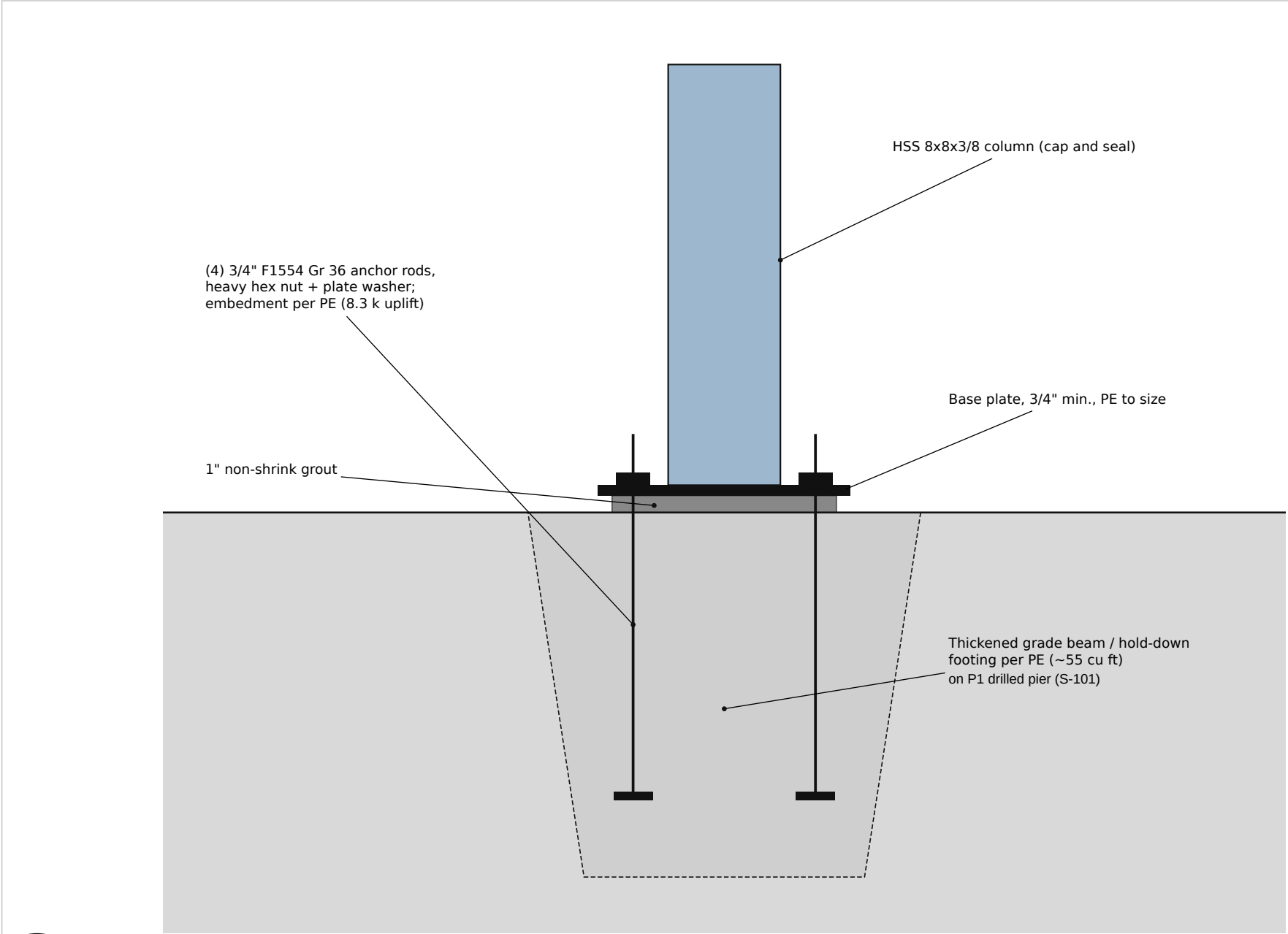


8 NORTH WALL (GRID A): PORTAL FRAMES AND WIND POSTS

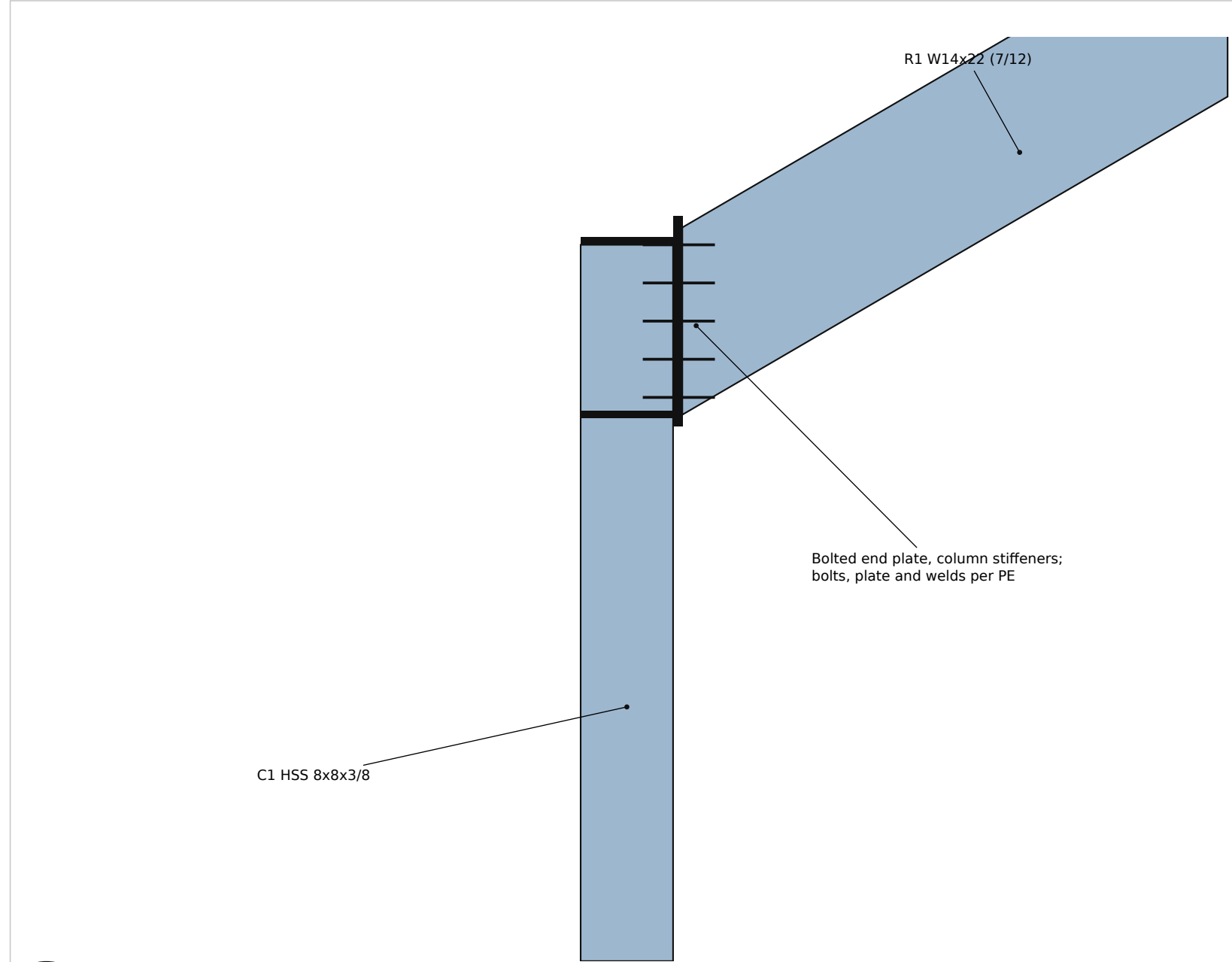
SCALE: 3/16" = 1'-0"

FRAME NOTES

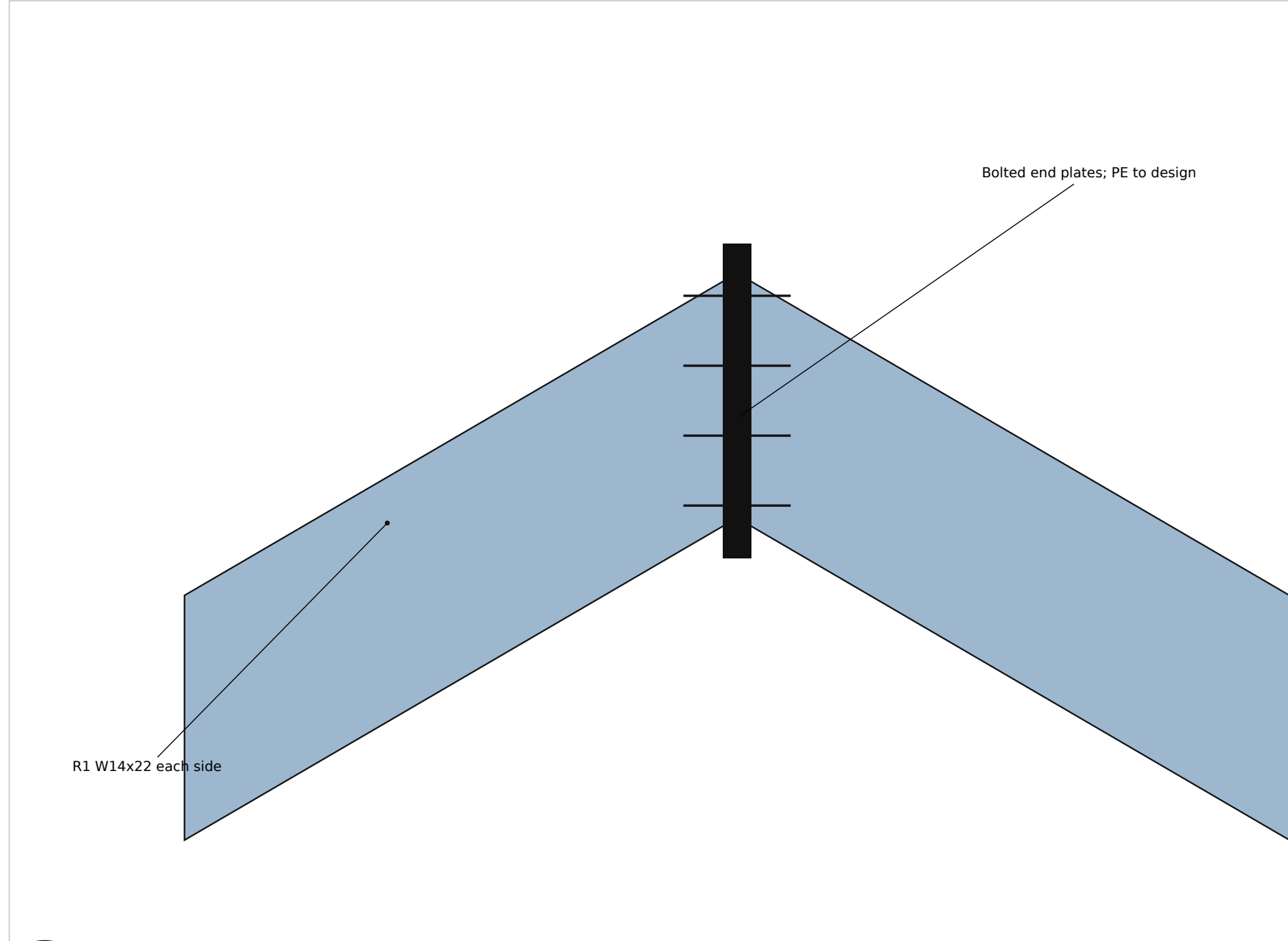
- Frame analysis: 140 mph, Exposure C, partially enclosed (GCp1 ±0.55). Utilization: column 0.69, rafter 0.72, porch 0.48, suite 0.58, header 0.37, portal 0.80, grid 2 frame 0.35, grid 7 corner columns 0.52. Sway 1.3" at 0.6W.
- Max net uplift per main column 8.3 k (ASD); anchorage and footings on P1 drilled piers per S-101.
- Knee and ridge: bolted end-plate moment connections preferred; PE to design.
- Partitions attached to frames need slip joints at the top (frame sway).



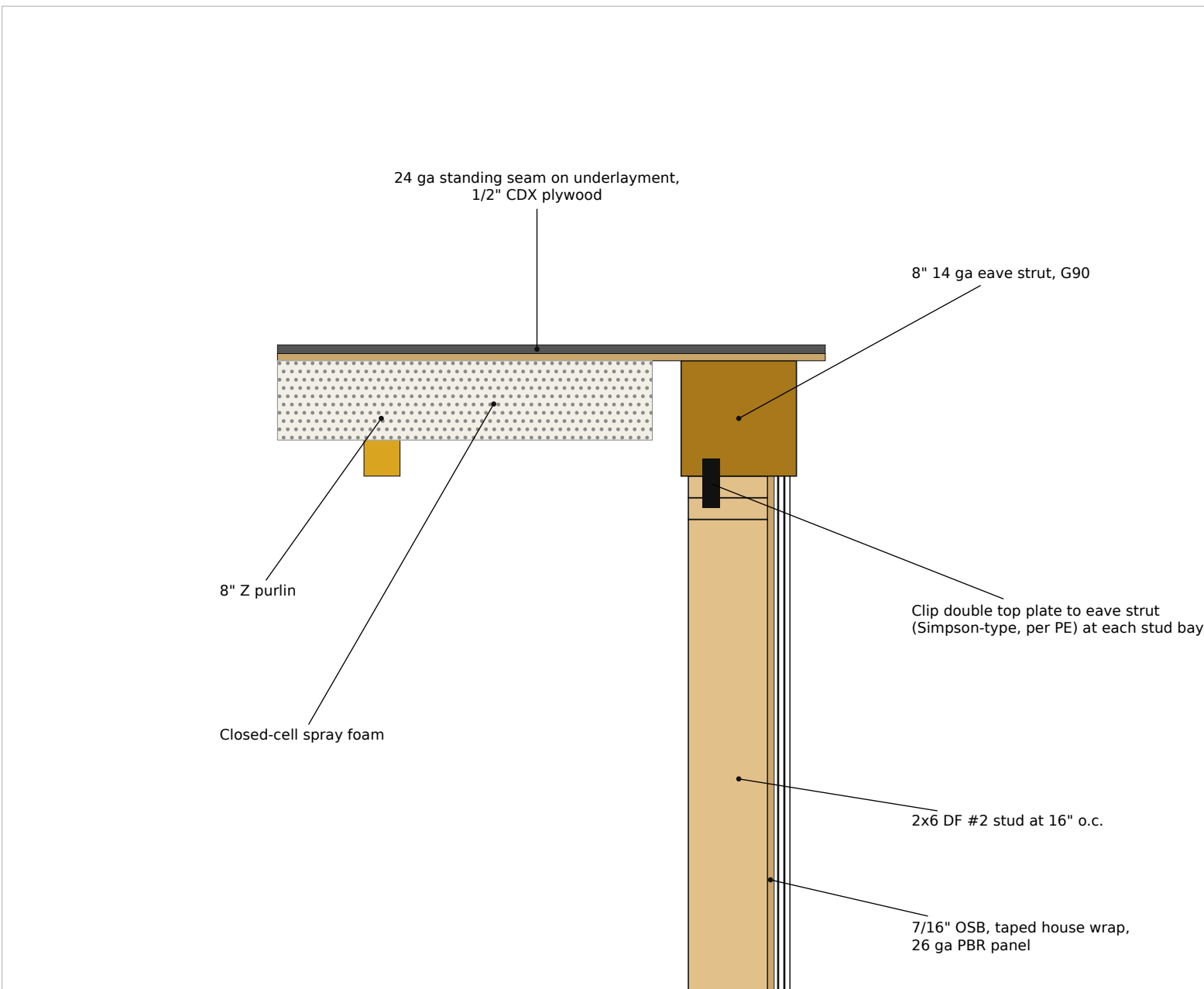
1 COLUMN BASE PLATE (BP-1)
SCALE: NTS



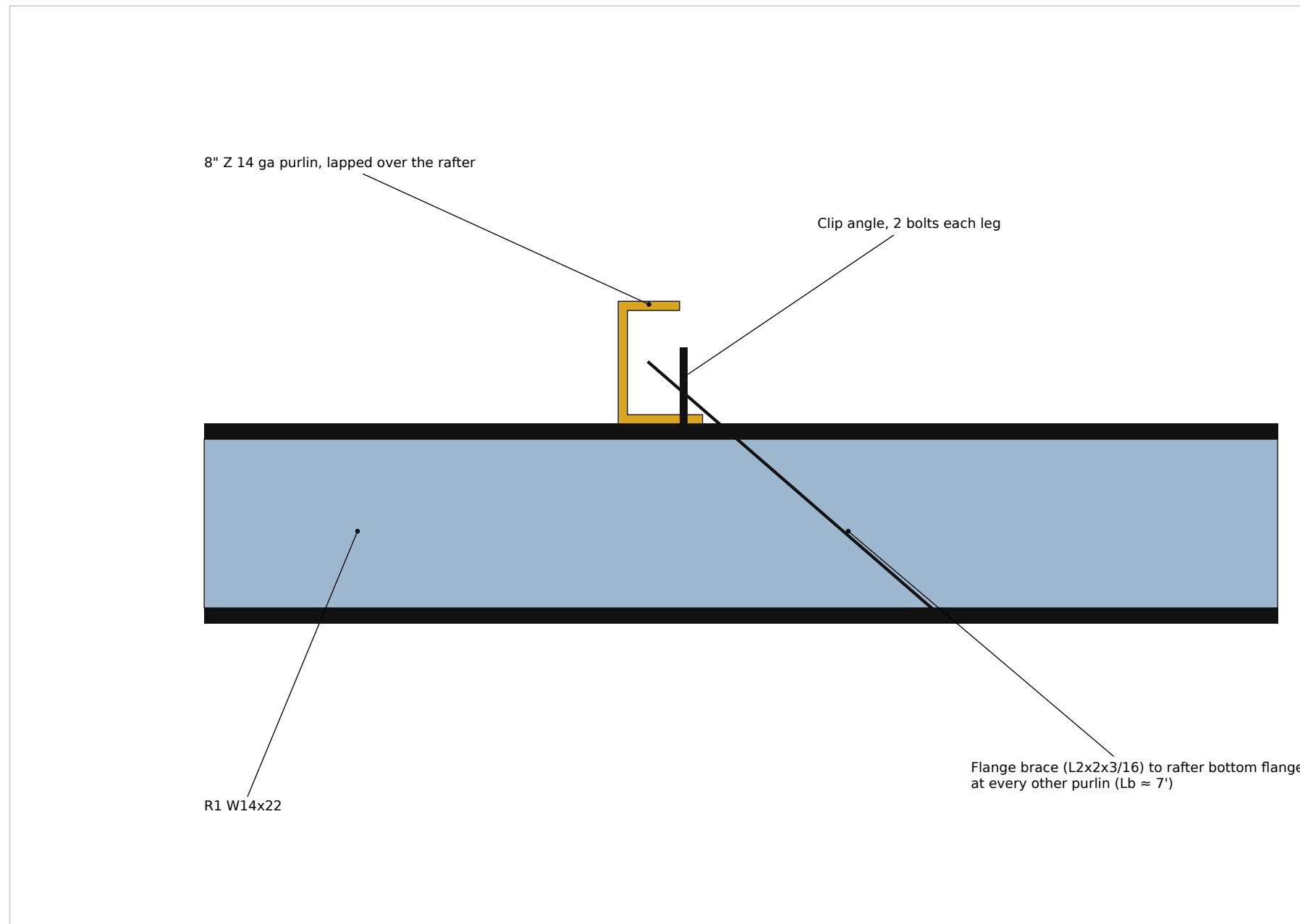
2 KNEE MOMENT CONNECTION
SCALE: NTS



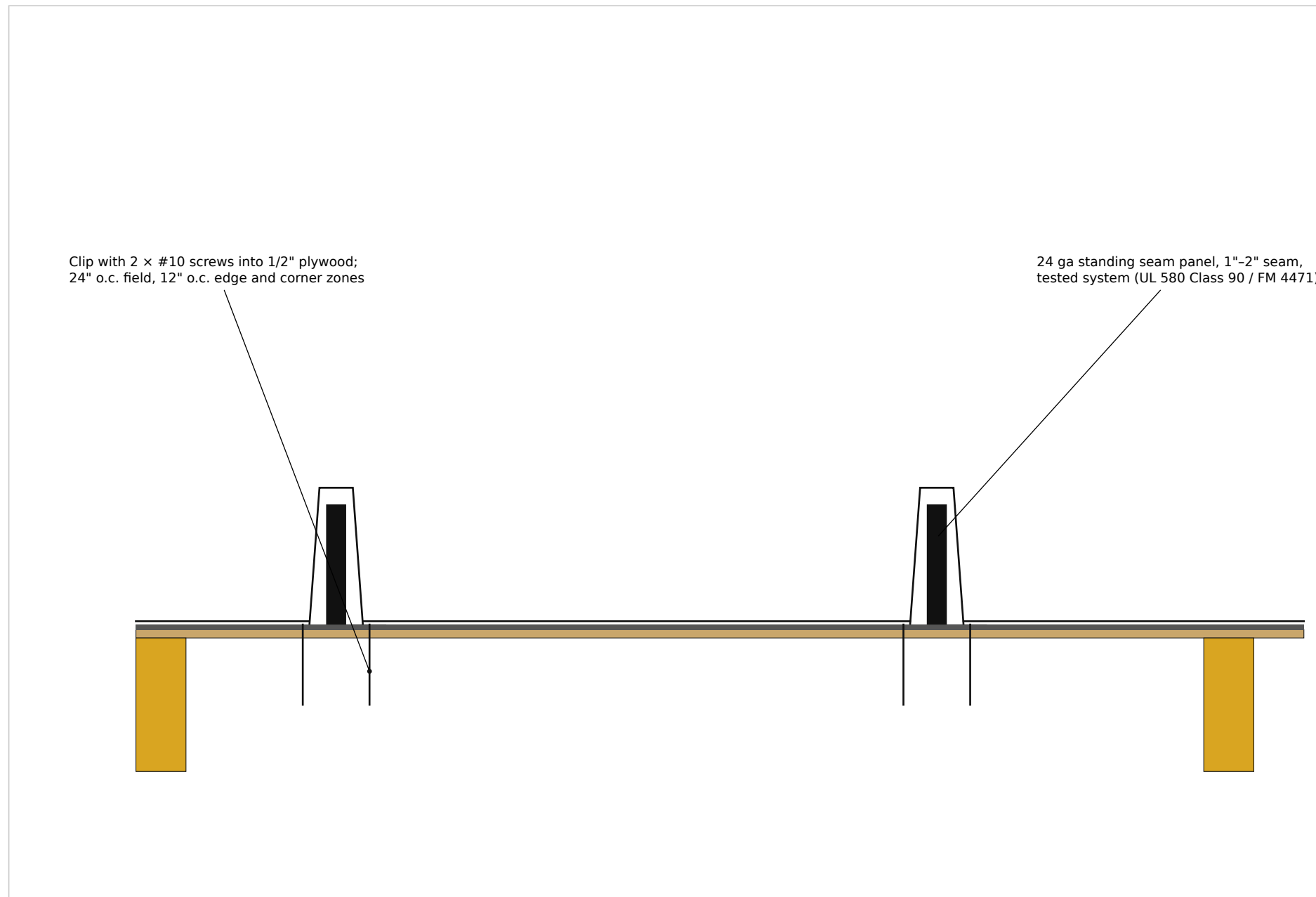
3 RIDGE SPLICE
SCALE: NTS



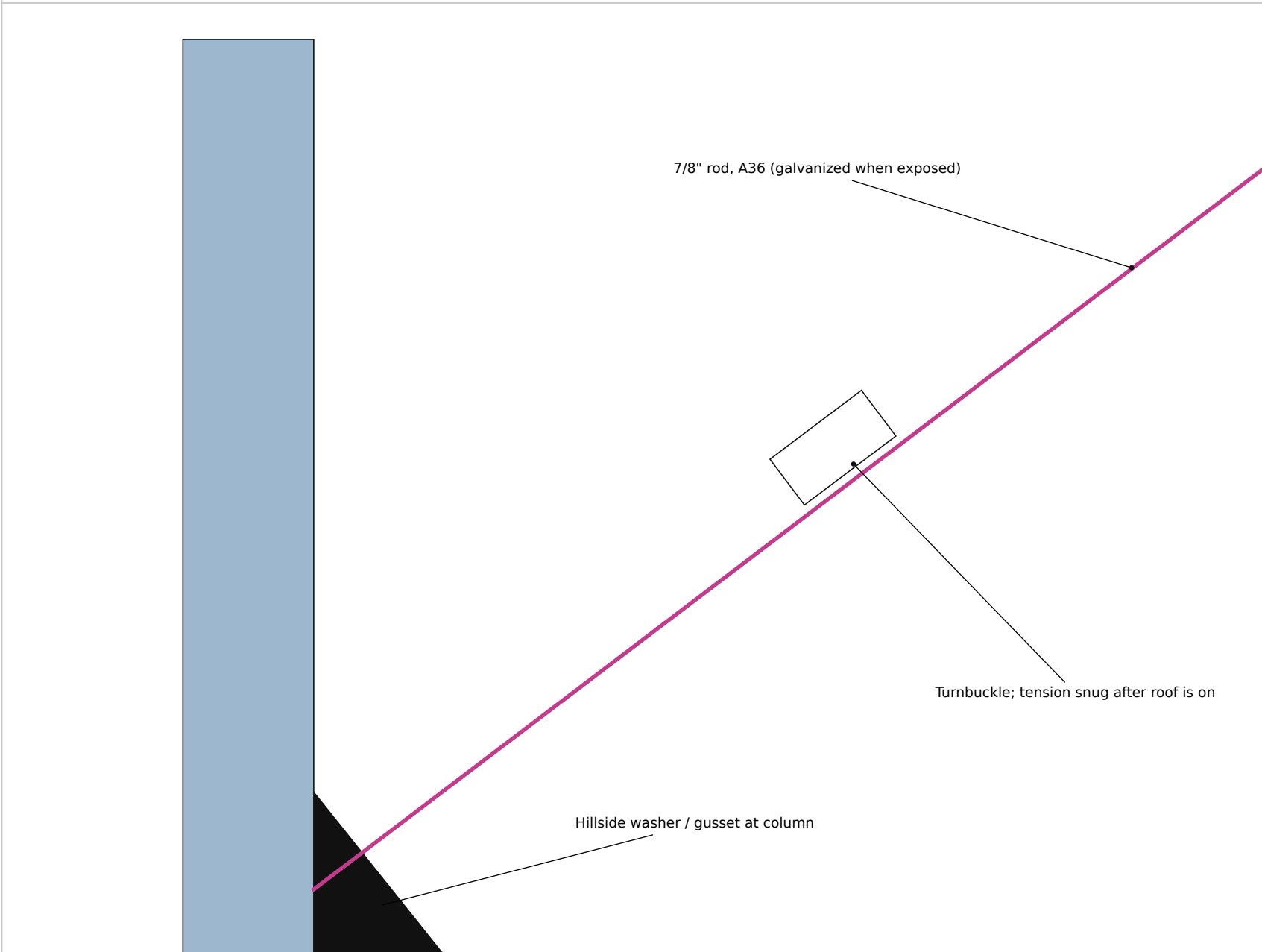
4 EAVE: WOOD WALL TO STEEL
SCALE: NTS



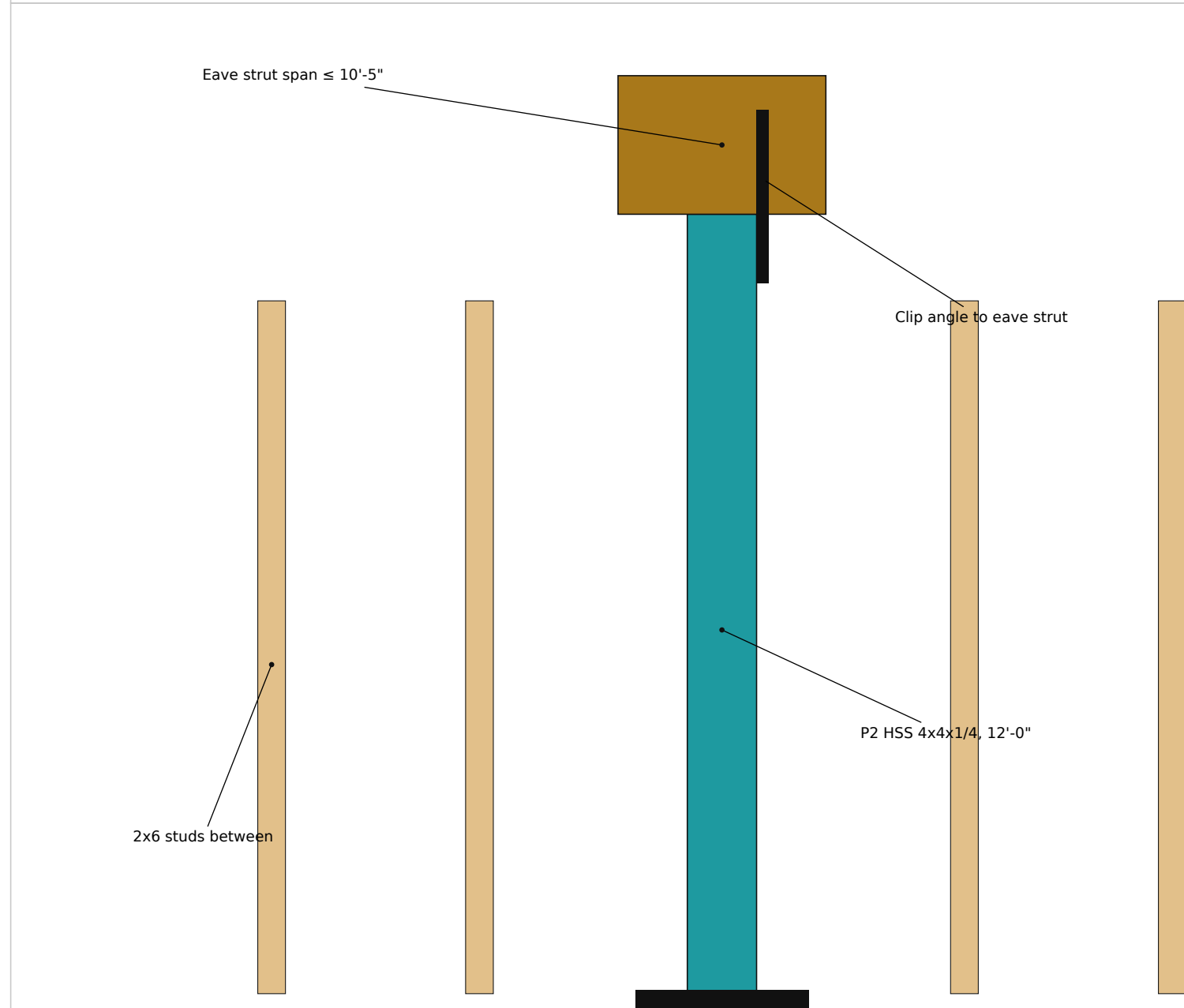
5 PURLIN AND FLANGE BRACE
SCALE: NTS



6 STANDING SEAM CLIP
SCALE: NTS



7 ROD X-BRACE END
SCALE: NTS



8 MID-BAY WIND POST
SCALE: NTS

CONNECTION NOTES

- Details are schematic and show intent only. The Texas PE designs every connection: bolts, welds, plates, embedments and fastener schedules.
- Structural steel: HSS ASTM A500 Gr C; W shapes A992; plates and angles A36; anchor rods F1554 Gr 36; high-strength bolts A325.
- Cold-formed purlins, girts and eave struts: G90 galvanized (ASTM A653).
- Fasteners in contact with Galvalume: panel maker's coated or stainless; no copper.

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PEREZ RESIDENCE

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ENGINEER OF RECORD (TEXAS PE)

Seal, signature
and date

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REVISIONS

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1	10/08/2026	Address; fireplace removed; vaults; BOM file renamed
2	10/08/2026	Pier note, BP-1

PREPARED FROM

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Septic: K&B Tech OSSF plan, 08-14-26.

SHEET TITLE

TYPICAL CONNECTION DETAILS

NOT TO SCALE

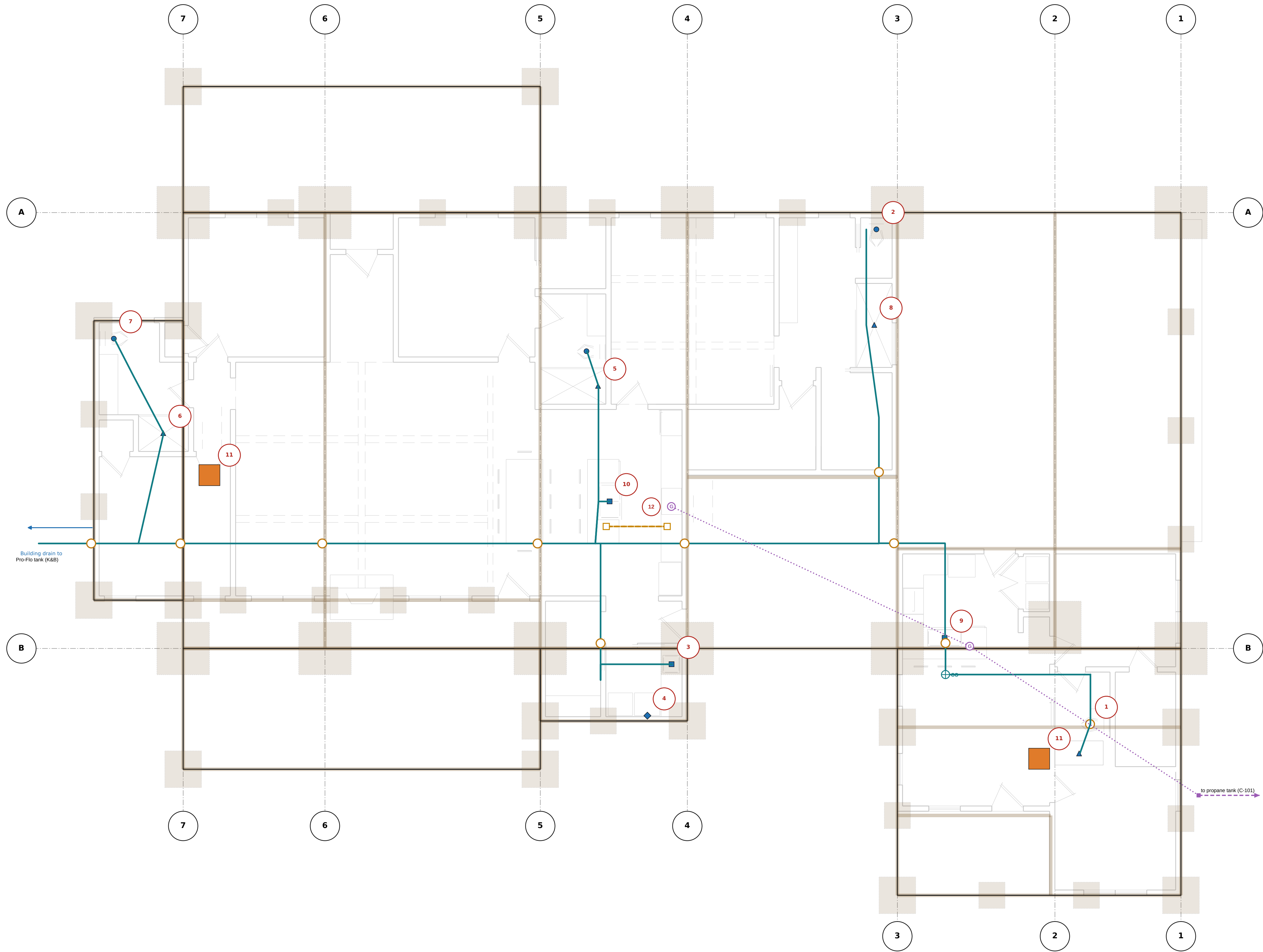
DATE
10/05/2026

DRAWN
Owner / Claude

SHEET

S-501

Preliminary design. The Texas PE's sealed
drawings and calculations govern.



COORDINATION ITEMS

- 1

Mom's toilet sits on the suite frame-line beam (GB-2 at 49'). Move the toilet about 1'-0" north (flange near 47'-8") or have the PE detail a sleeved block-out. Her curbless shower recess (from 50'-3") must also stay off this beam.
Owner / PE
- 2

Master toilet flange falls inside footing F1 at grid 3/A (footing to 29'-6"). Shift the flange 1' west or have the PE make F1 eccentric. It backs onto the north exterior wall: feed it from the floor.
PE / plumber
- 3

Laundry sink drain falls inside footing F1 at grid 4/B (to 44'-0" south). Set the sink south of 44'-6" or route the drain clear. Wall faces the open breezeway: floor-fed supply.
Plumber
- 4

Washer box on the shed rear wall sits over GB-1 and in an exterior wall. Put the box on the laundry/pantry partition (55'-0") next to the washer.
Owner / plumber
- 5

Bath 2 shower: drain at the east end (=55'-6"). The west end sits over the grid 5 beam (61').
Plumber
- 6

Bath 3 shower straddles the grid 7 beam (95'). Drain at least 1'-6" west of 95'; keep the slab recess clear of the beam.
Plumber / PE
- 7

Bath 3 toilet flange is only ~2" outside footing F2 at the bump corner and backs onto the exterior wall. Shift it ~4" east (flange 6"+ clear of the footing); floor-fed supply.
Plumber
- 8

Master shower is against the carport wall (exterior) and 1'-6" from the grid 3 beam. Valve on an interior partition; drain at least 1'-6" west of 27'.
Plumber
- 9

Mom's kitchen sink drain is 1'-0" from the beam at 41'-6". Keep the trap arm north of 40'-0"; loop vent or AAV at the peninsula.
Plumber
- 10

Kitchen island sink: island vent (loop vent or AAV where the county allows).
Plumber
- 11

Attic water heaters: framed platform for ~600 lb, pan and T&P drains to daylight, access per A-601. The main heater is ~70' from the master bath: add a recirculation pump.
PE / plumber

PLUMBING NOTES

1. Drains cross grade beams only horizontally, in sleeves two pipe sizes larger, at mid-depth; no vertical penetrations through beams or footings.
2. The building drain leaves the west end to the Pro-Flo tank (K&B site plan), about 10' outside the slab. Mom's suite is combined with the main house drain: toilet and shower tie in through the kitchen-sink branch at the start of the main run (cleanout 22'-2" / 44'-6"; sleeved crossings at 41'-6" and 27').
3. All water supply lines in interior walls or up through the slab in insulated chases; none in exterior walls (A-601).
4. Sleeve every slab penetration and fit slab-exit collars (expansive clay). Plumbing rough-in survey before the pour, like the anchor rods.
5. Curbless showers: slab recesses per the PE, kept off beams and footings.
6. Fixture locations follow the Quality First layout; the plumber confirms each drain against this sheet before trenching.

ELECTRICAL AND PIER COORDINATION

- 12

Kitchen island power: cast in (2) 1" PVC conduits under the slab from the base of the east counter wall (about 48'-7") to the island base (about 54'-5"). One for power, one spare. Keep 2'-0" off the sink drain and 1'-7" off the main drain. Long-sweep elbows, sleeve and collar where they leave the slab, capped and marked at both ends.
Electrician / plumber

Piers: 13 drilled piers P1 (18" x 15') are drilled under the F1 footings at the main columns before rough-in (S-101). Keep drains, sleeves and conduit 3'-0" clear of every pier shaft.

Septic power: the Pro-Flo unit runs on 110 V. Flow Co. connects at the unit but cannot add lines or breakers in the house, so the electrician runs a dedicated circuit and conduit from the panel to the tank and leaves a wire for the unit's AC disconnect (estimate EST-0015). Confirm the circuit and any alarm wiring on the K&B plan.

Propane (LP-gas): the 500 gal tank is about 26' east of Mom's suite (see C-101). A buried 3/4" PE line, 18" deep with tracer wire, enters at an anodeless riser outside the suite east wall, between the F2 footings. Gas piping then runs overhead (attic and walks) to the suite range and on to the kitchen range, with a shutoff at each. No gas pipe goes under the slab or through a beam. The propane contractor sizes, tests and approves the system; the two ranges are the only gas appliances assumed.

1

PLUMBING COORDINATION PLAN

SCALE: 3/16" = 1'-0"

- Toilet flange

■ Sink drain

◆ Washer box

▲ Shower / tub drain

— Sanitary drain route, schematic only
- ▬ Grade beam (12" wide)

▬ Column footing F1 / F2 / F3

■ Water heater + air handler (attic)

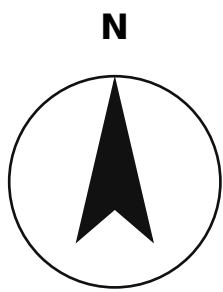
○ Coordination item (table)

○ Beam sleeve, PE/plumber to detail
- Electrical conduit, island power (stub-up squares)

⊕ Cleanout (CO)

- - - Propane (LP-gas) line, schematic; G = gas range

Sanitary only. Water and vent lines are not shown. Confirm pipe size, slope, invert, cleanouts and final route before construction.



PRELIMINARY
FOR PE REVIEW

NOT FOR
CONSTRUCTION

PEREZ RESIDENCE

Perez Ranch
43 Percy Howard Rd, Huntsville, TX
Walker County (unincorporated)

Owner: Carlos Perez

ENGINEER OF RECORD (TEXAS PE)



Name / firm / F-number:

REVISIONS

No.	Date	Description
0	10/05/2026	Preliminary for PE review
1	10/08/2026	Address; fireplace removed; vaults; BOM file renamed
2	10/08/2026	Island conduit; piers
3	10/08/2026	Drains combined; K&B
4	10/08/2026	Propane lines added

PREPARED FROM

Owner's 3D frame model and BOM
(Perez Ranch Material BOM.xlsx).

Room layout: Quality First Professional
Design Services plan set (rev. 03-25-26).

Geotech: SolisTEC KT26G035B.

Septic: K&B Tech OSSF plan, 08-14-26.

SHEET TITLE

PLUMBING
COORDINATION PLAN

SCALE: 3/16" = 1'-0"

DATE
10/05/2026

DRAWN
Owner / Claude

SHEET

P-101

Preliminary design. The Texas PE's sealed drawings and calculations govern.